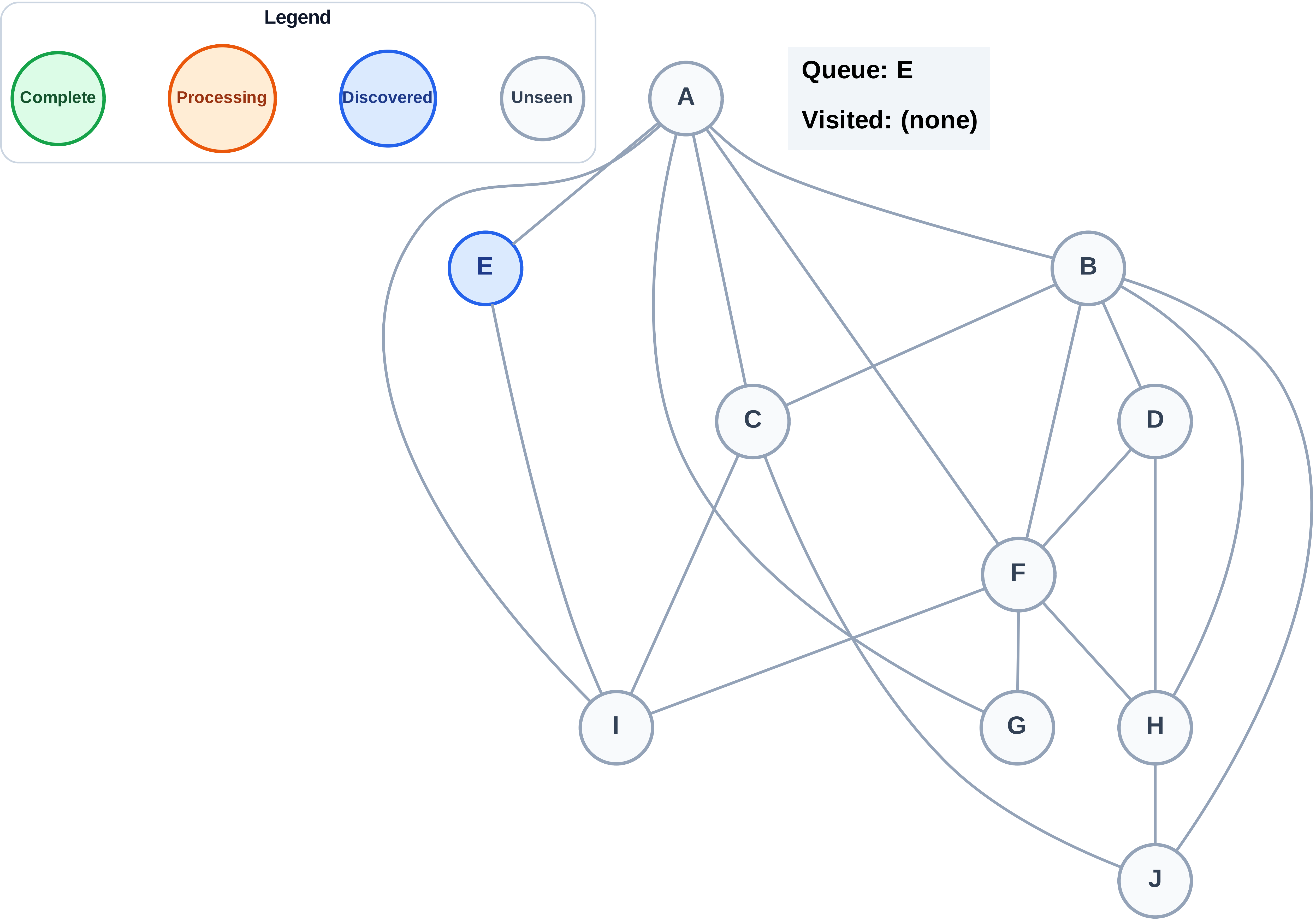
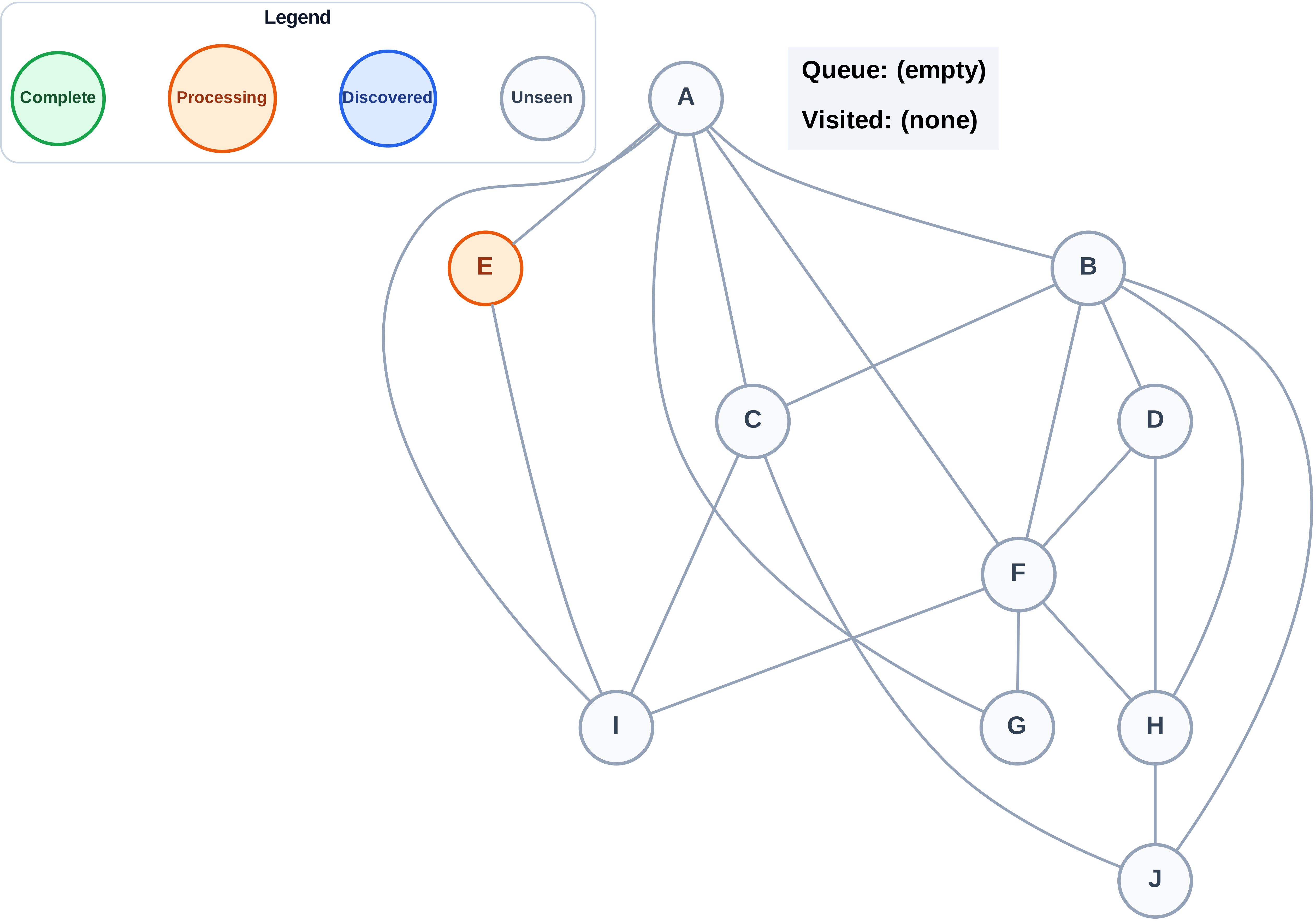
BFS Traversal

Step 1: Start at E



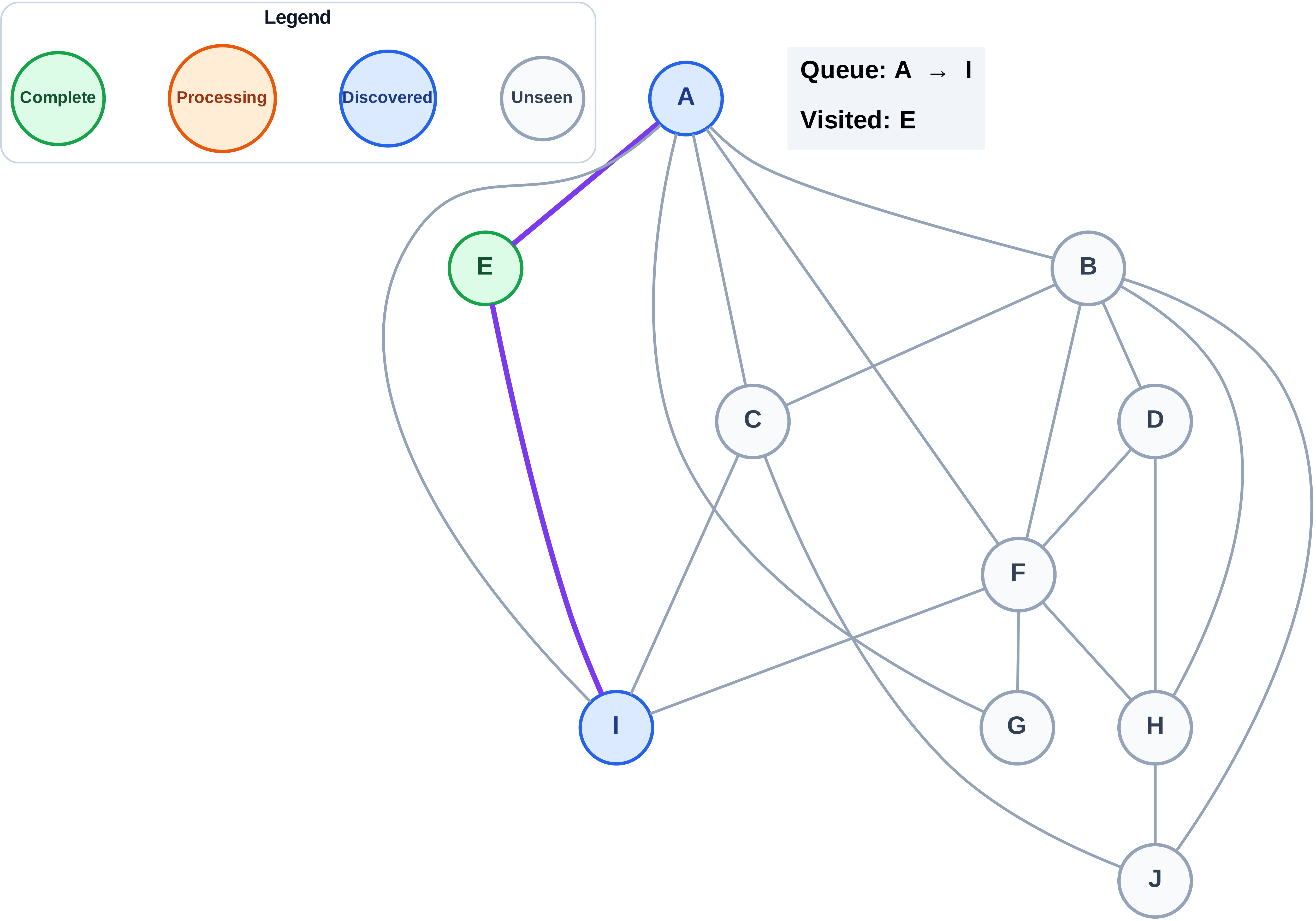
BFS Traversal

Step 2: Process node E



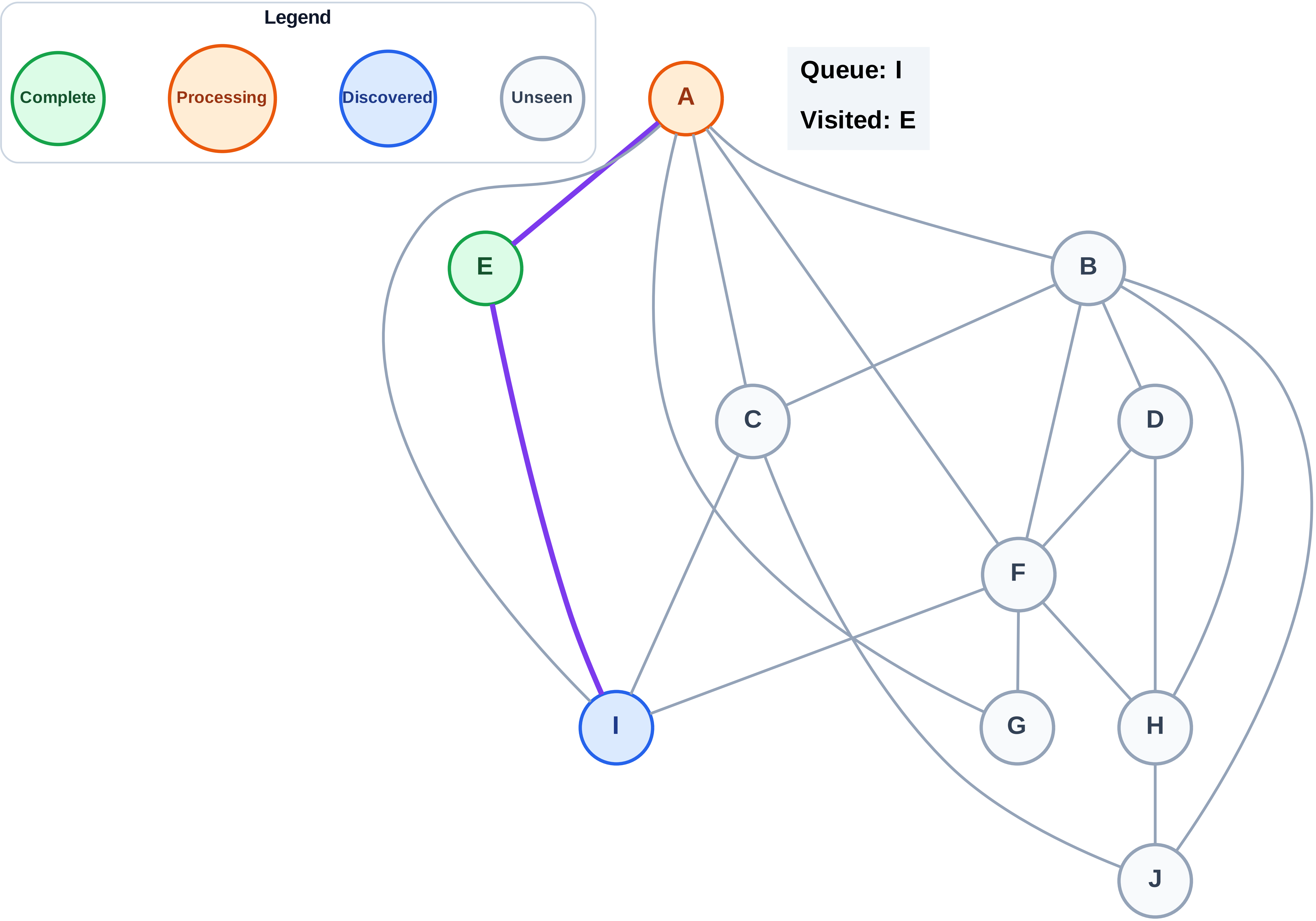
BFS Traversal

Step 3: Discover A, I from E



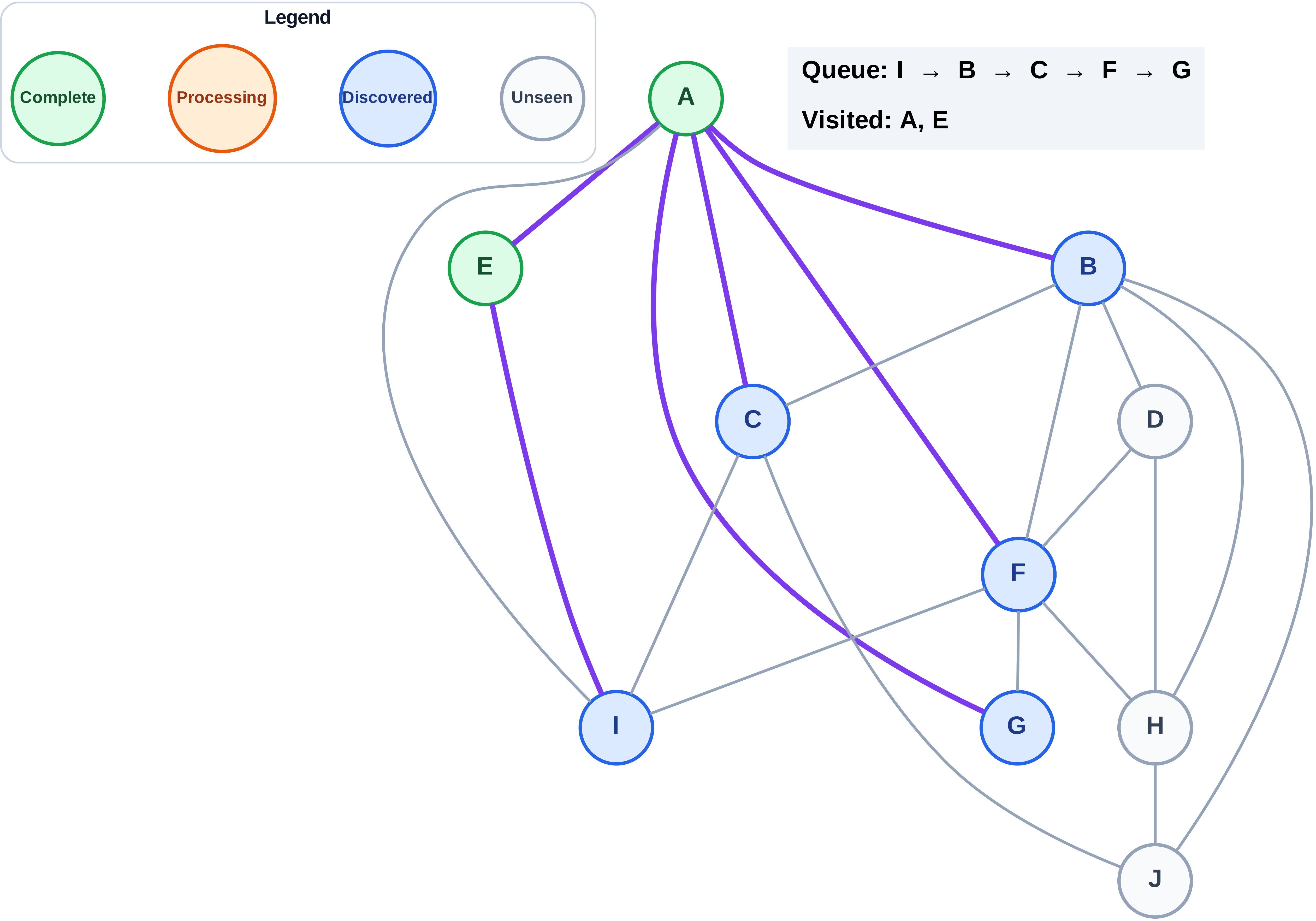
BFS Traversal

Step 4: Process node A



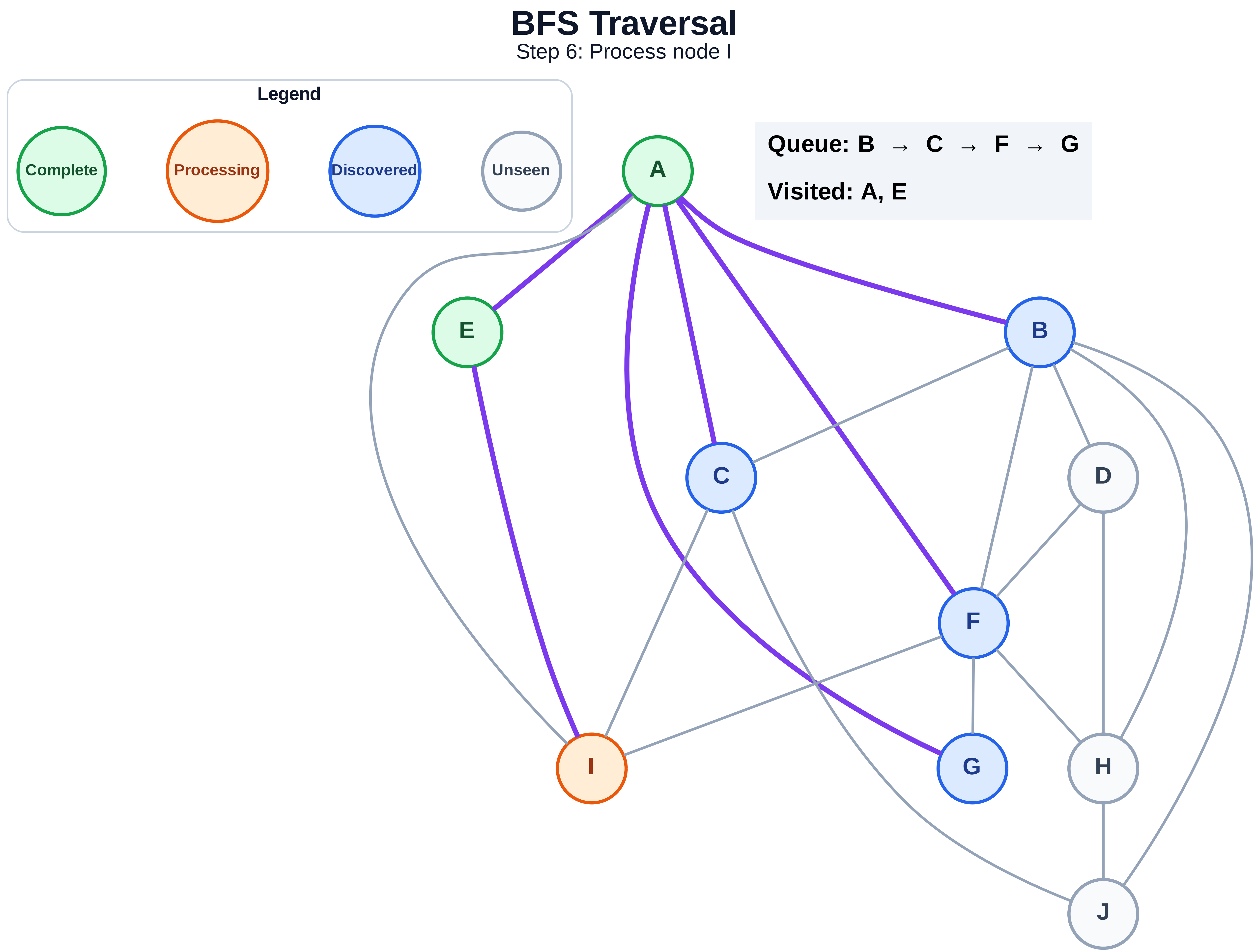
BFS Traversal

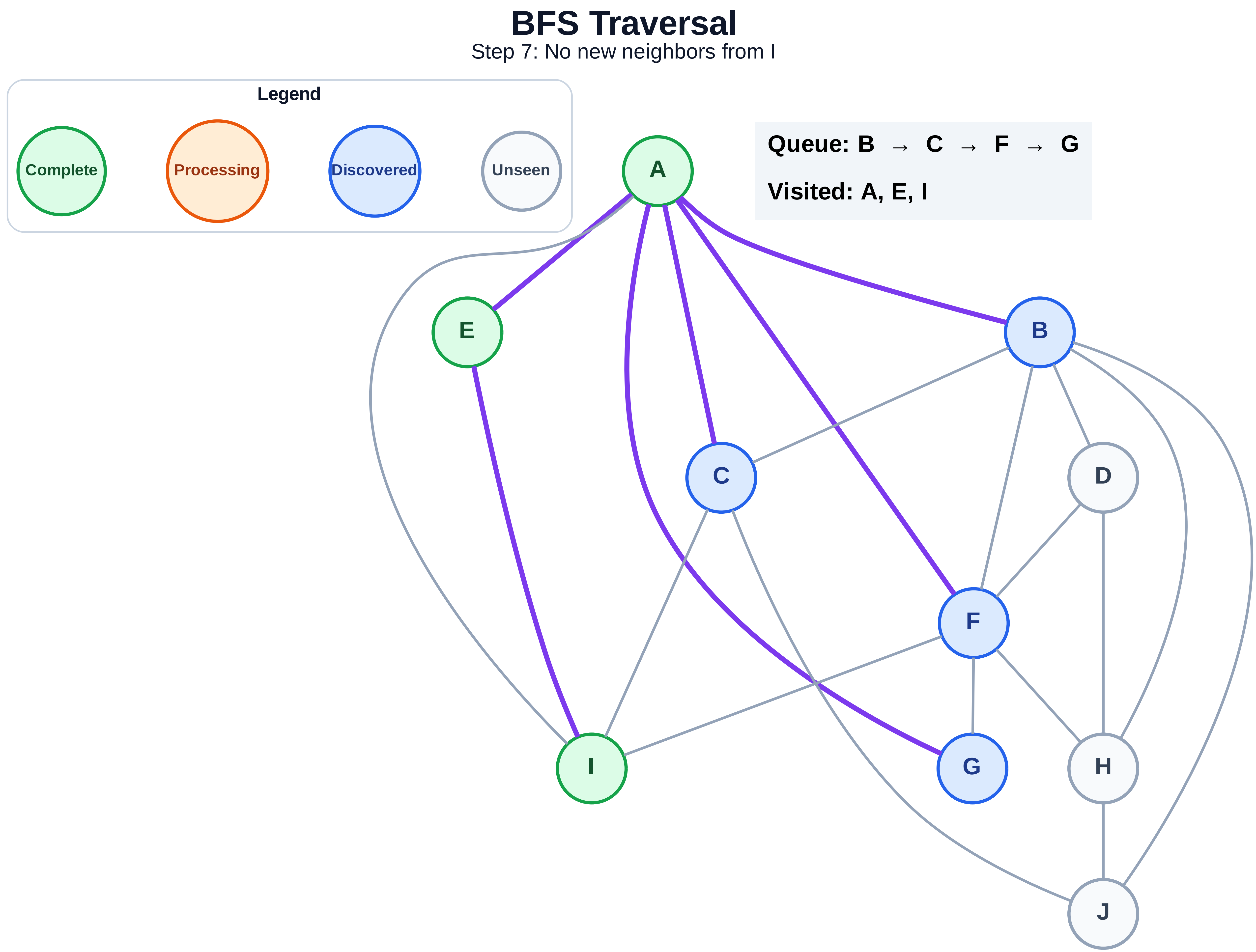
Step 5: Discover B, C, F, G from A



Queue: I → B → C → F → G

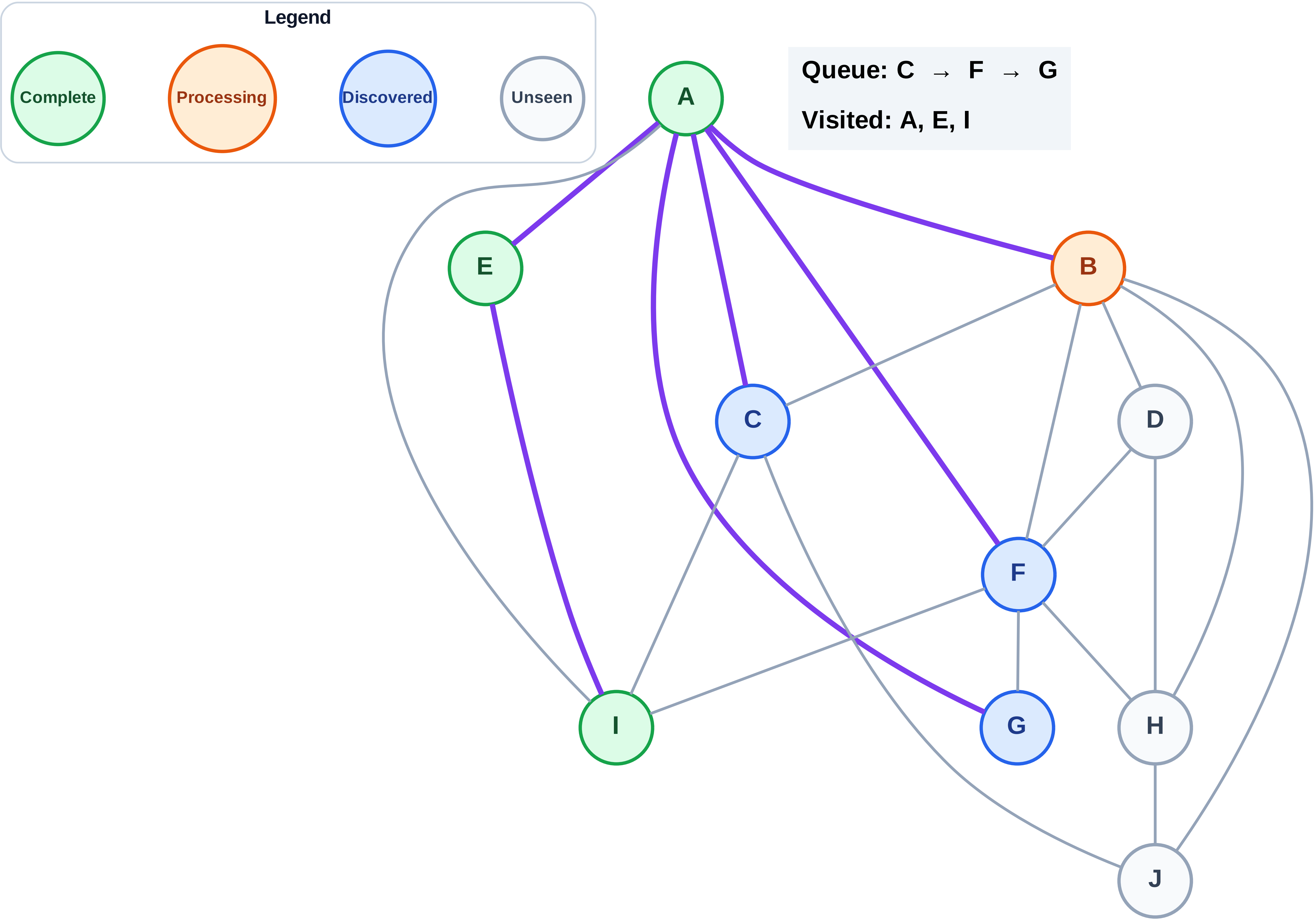
Visited: A, E


BFS Traversal
Step 6: Process node I
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → F → G
Visited: A, E
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 7: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → F → G
Visited: A, E, I
A
B
C
D
E
F
G
H
I
J

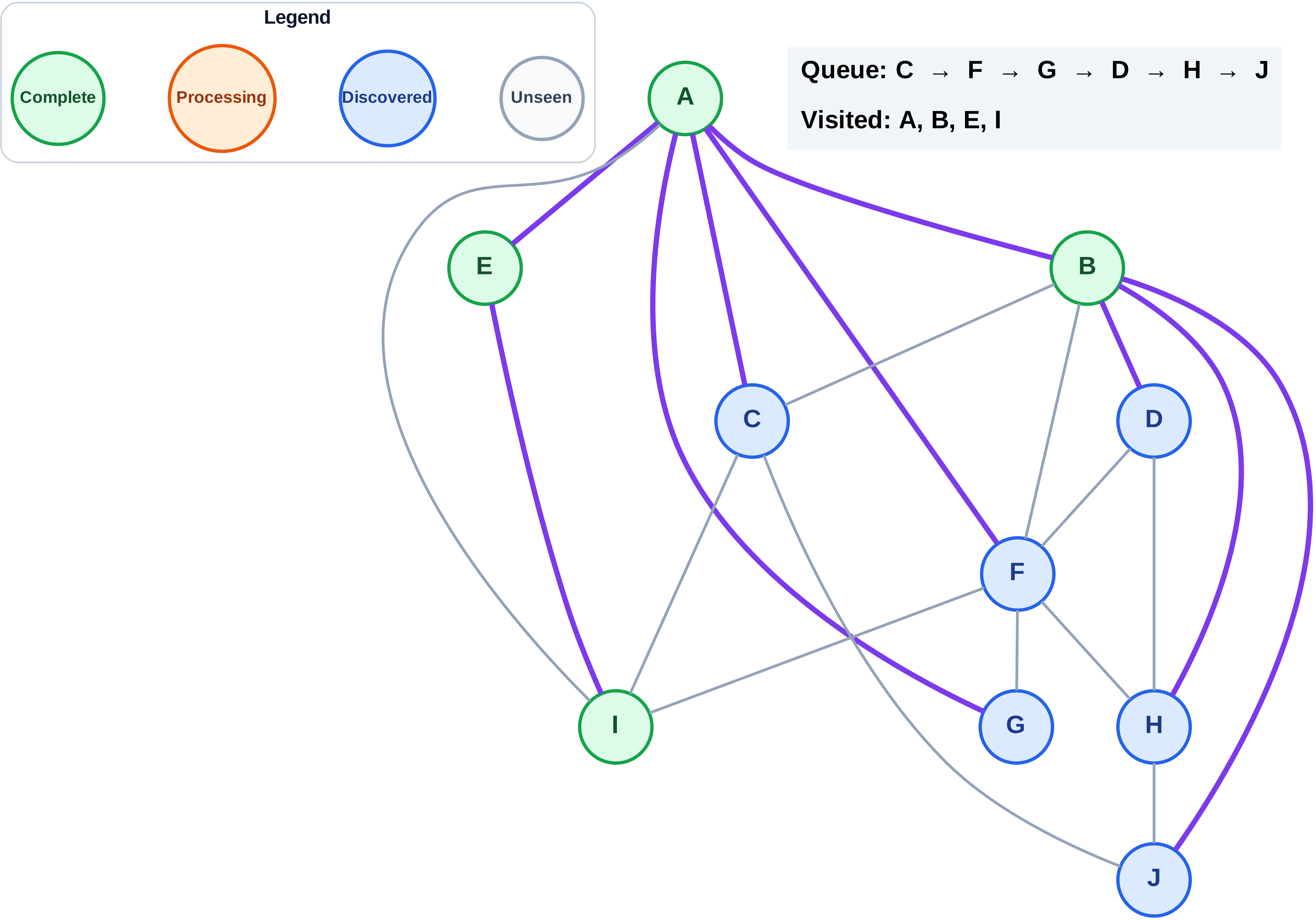
BFS Traversal

Step 8: Process node B



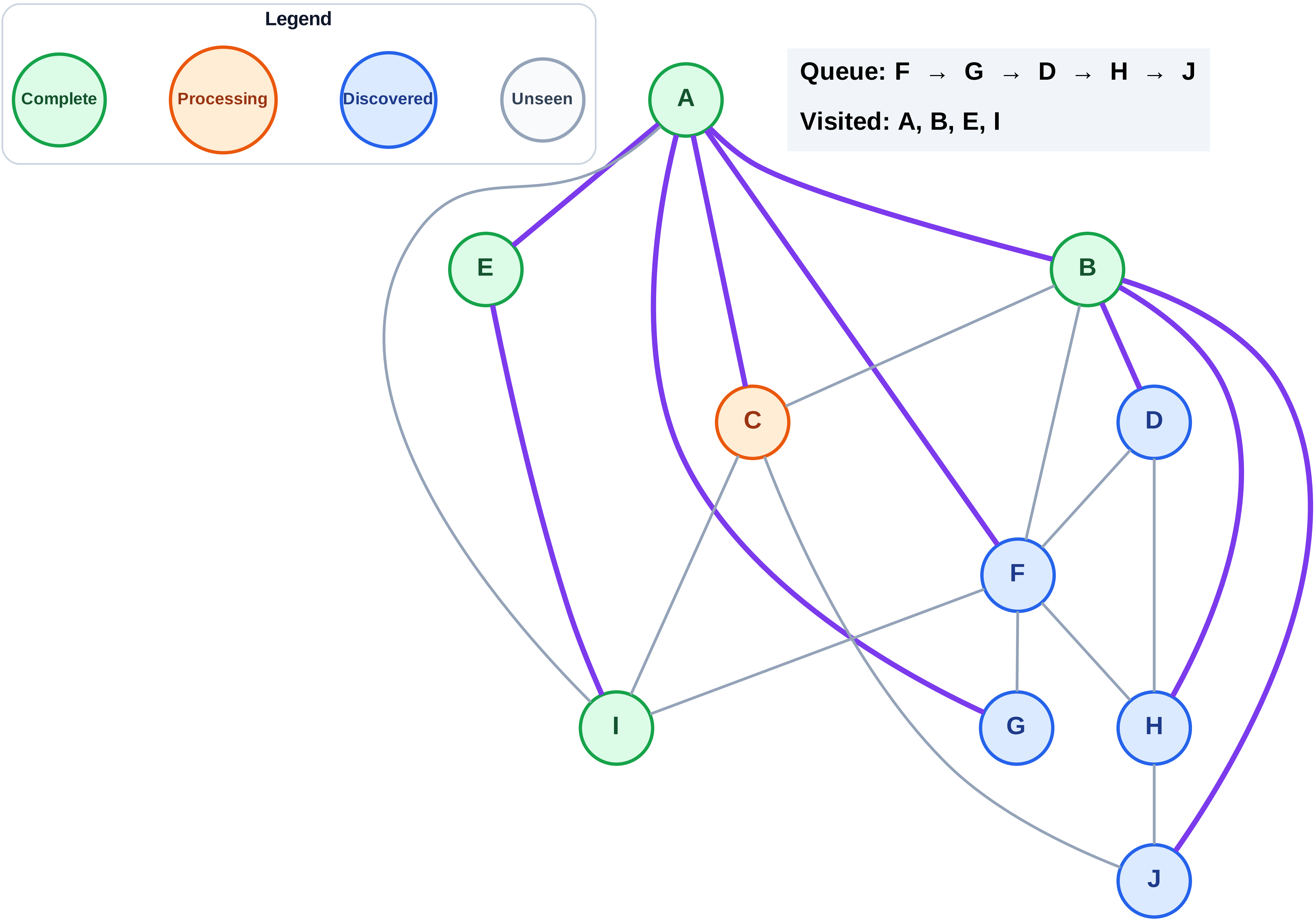
BFS Traversal

Step 9: Discover D, H, J from B



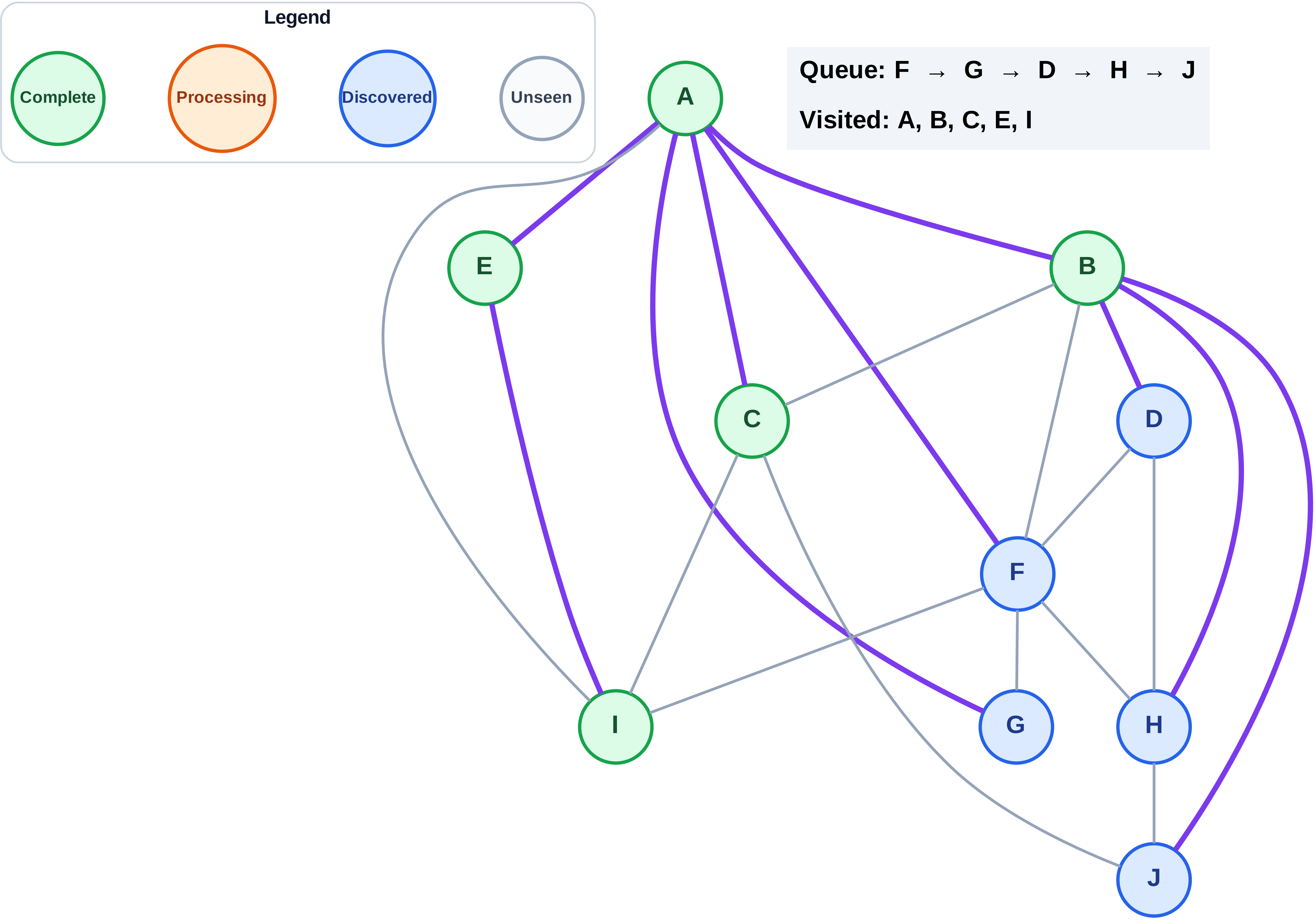
BFS Traversal

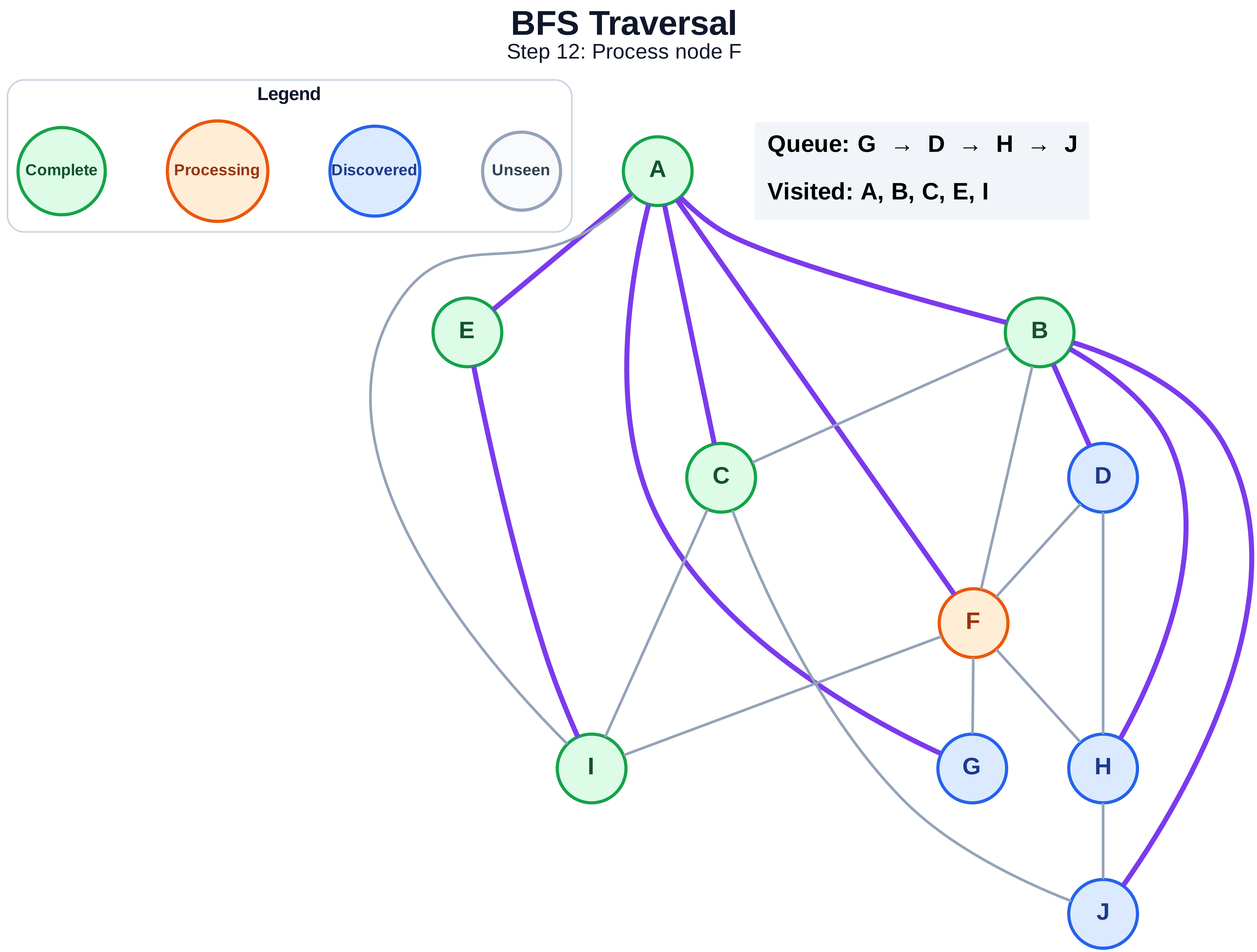
Step 10: Process node C



BFS Traversal

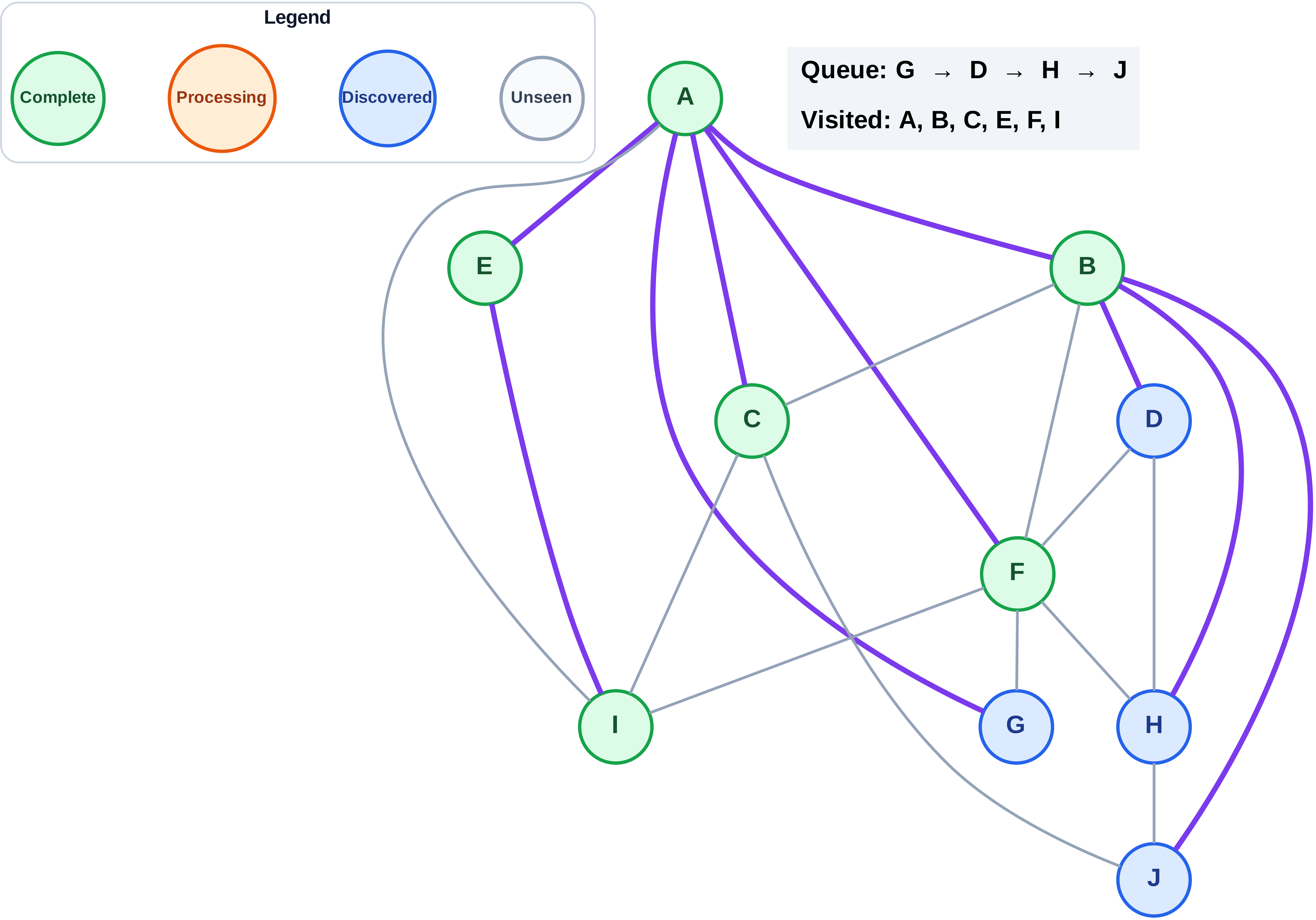
Step 11: No new neighbors from C




BFS Traversal
Step 12: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → D → H → J
Visited: A, B, C, E, I
A
B
C
D
E
F
G
H
I
J

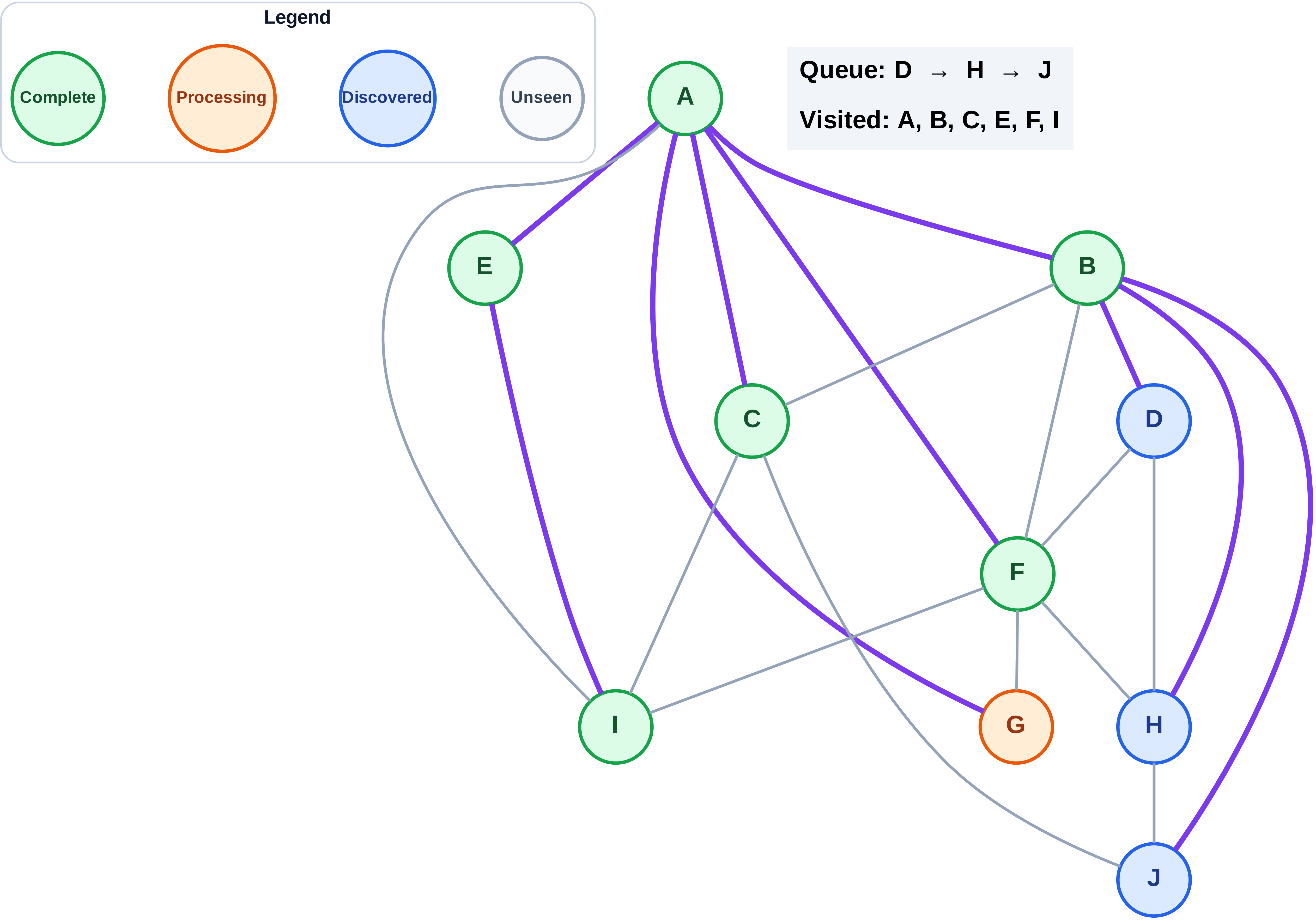
BFS Traversal

Step 13: No new neighbors from F



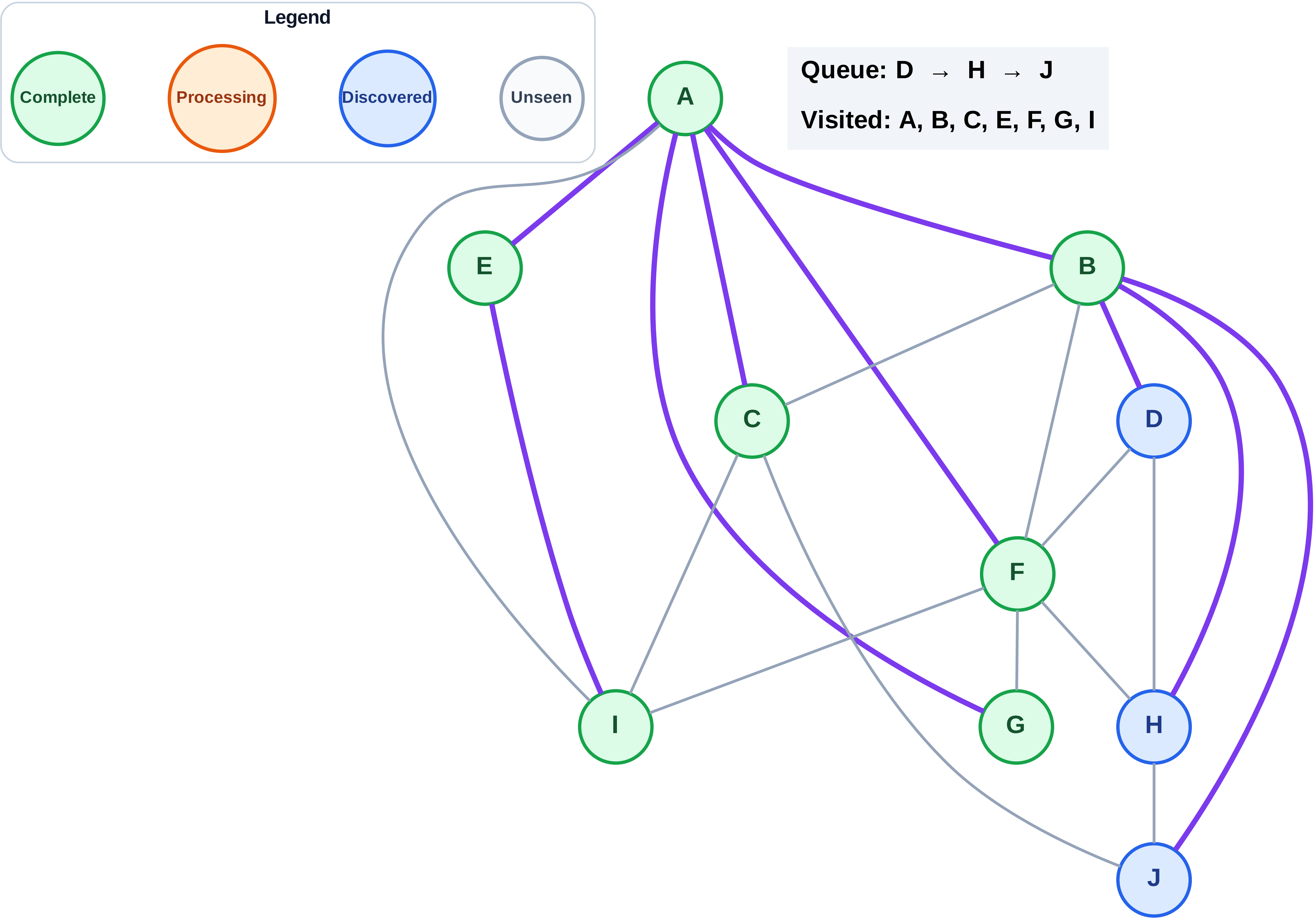
BFS Traversal

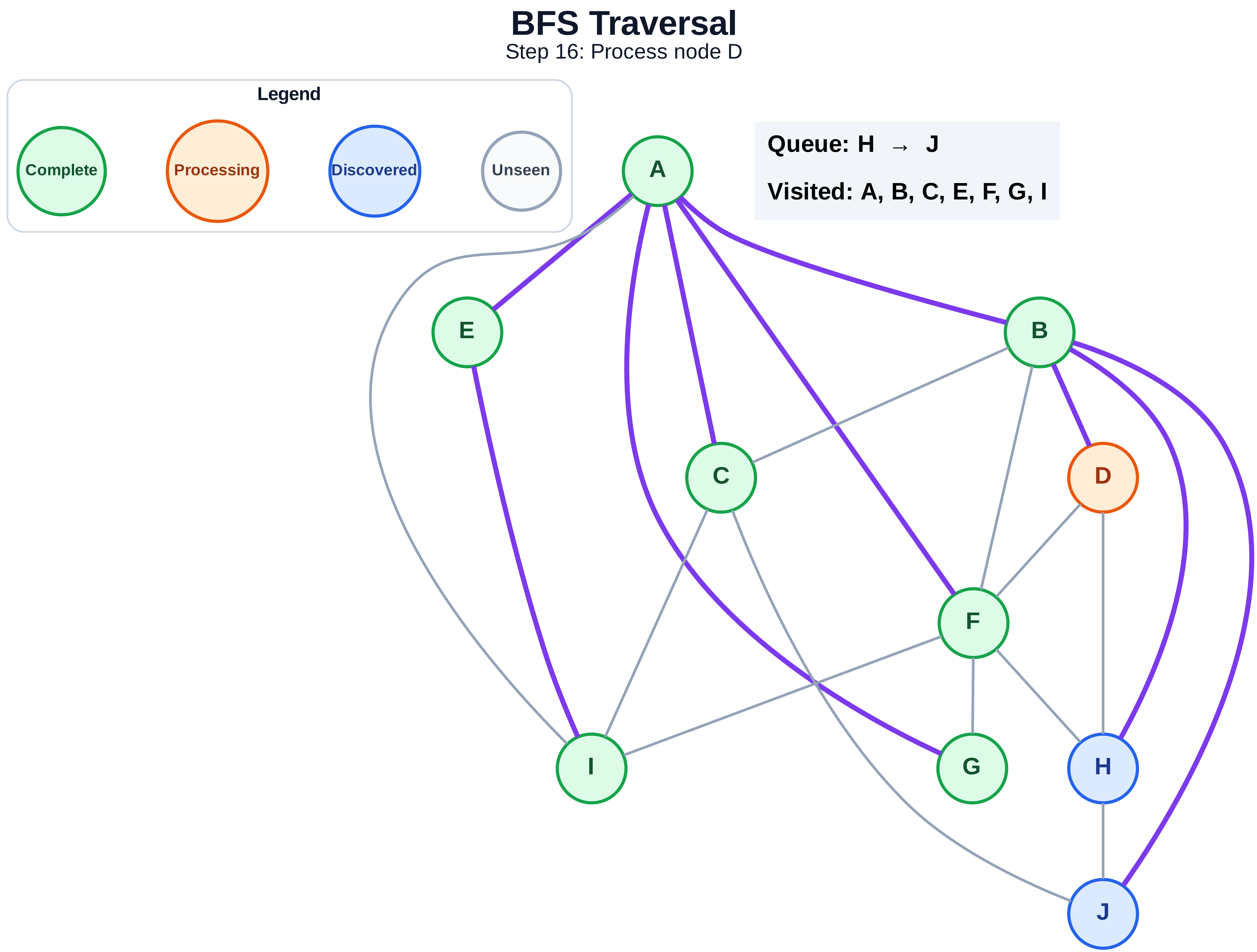
Step 14: Process node G

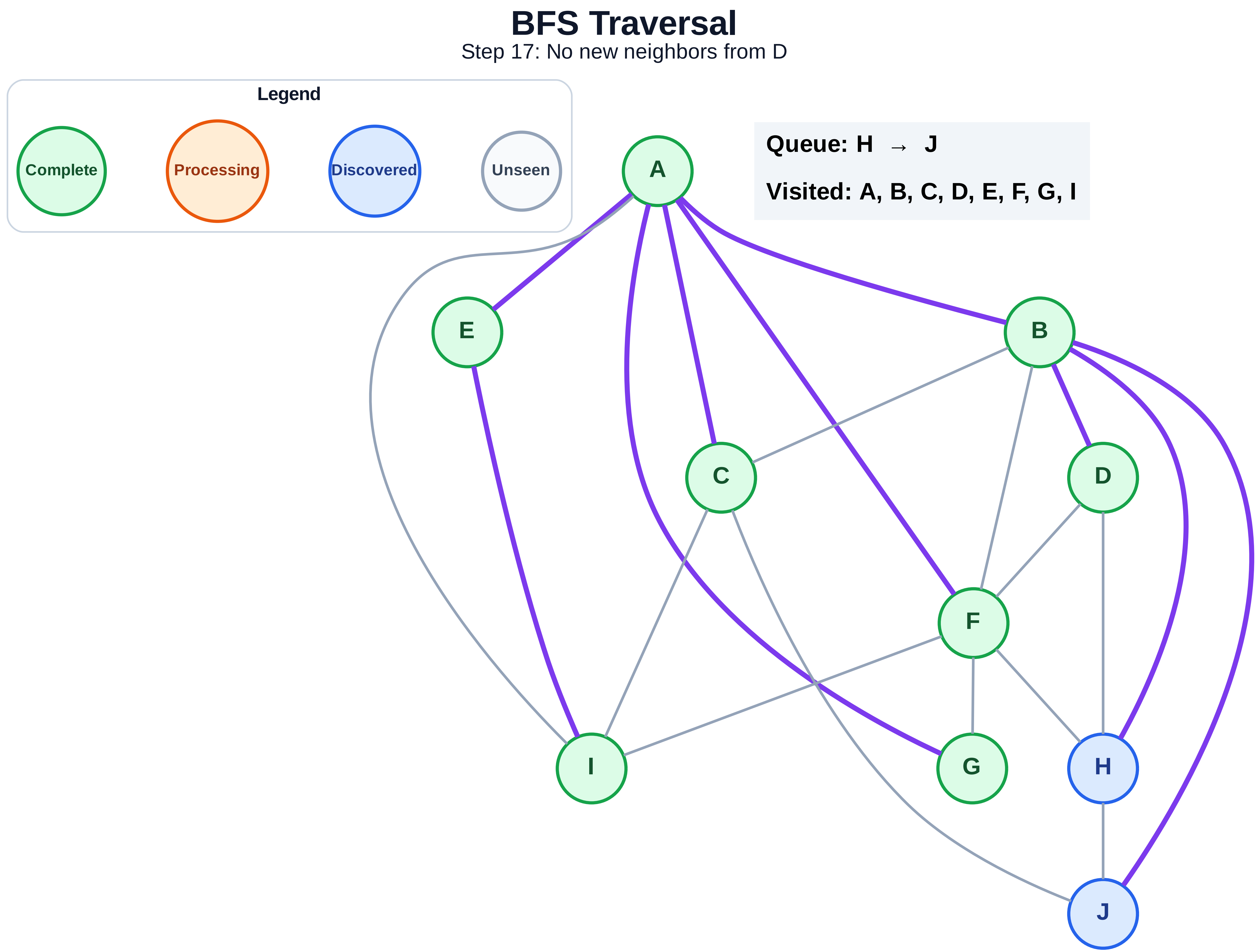


BFS Traversal

Step 15: No new neighbors from G

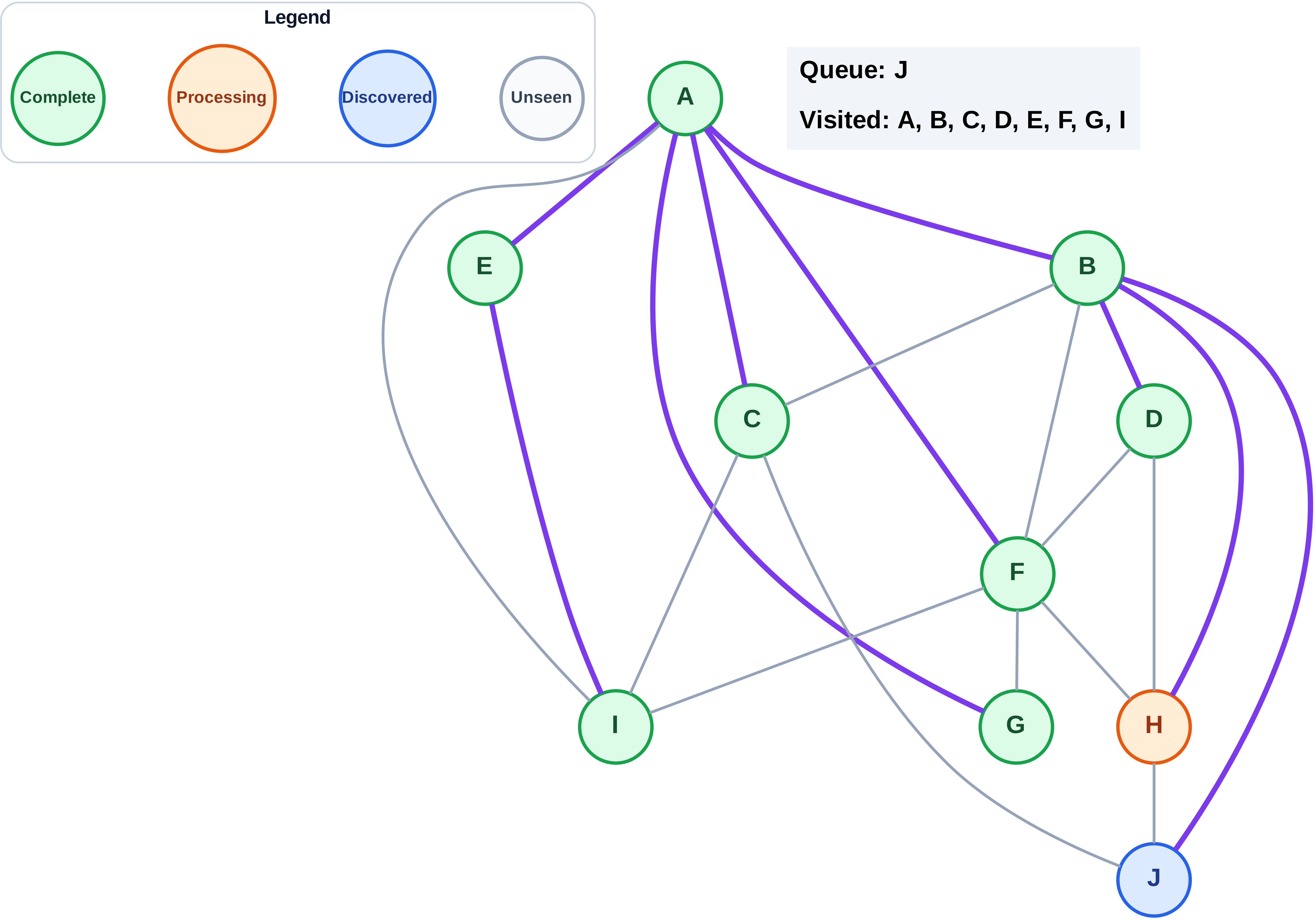



BFS Traversal
Step 16: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: H → J
Visited: A, B, C, E, F, G, I
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 17: No new neighbors from D
Legend
Complete
Processing
Discovered
Unseen
Queue: H → J
Visited: A, B, C, D, E, F, G, I
A
B
C
D
E
F
G
H
I
J

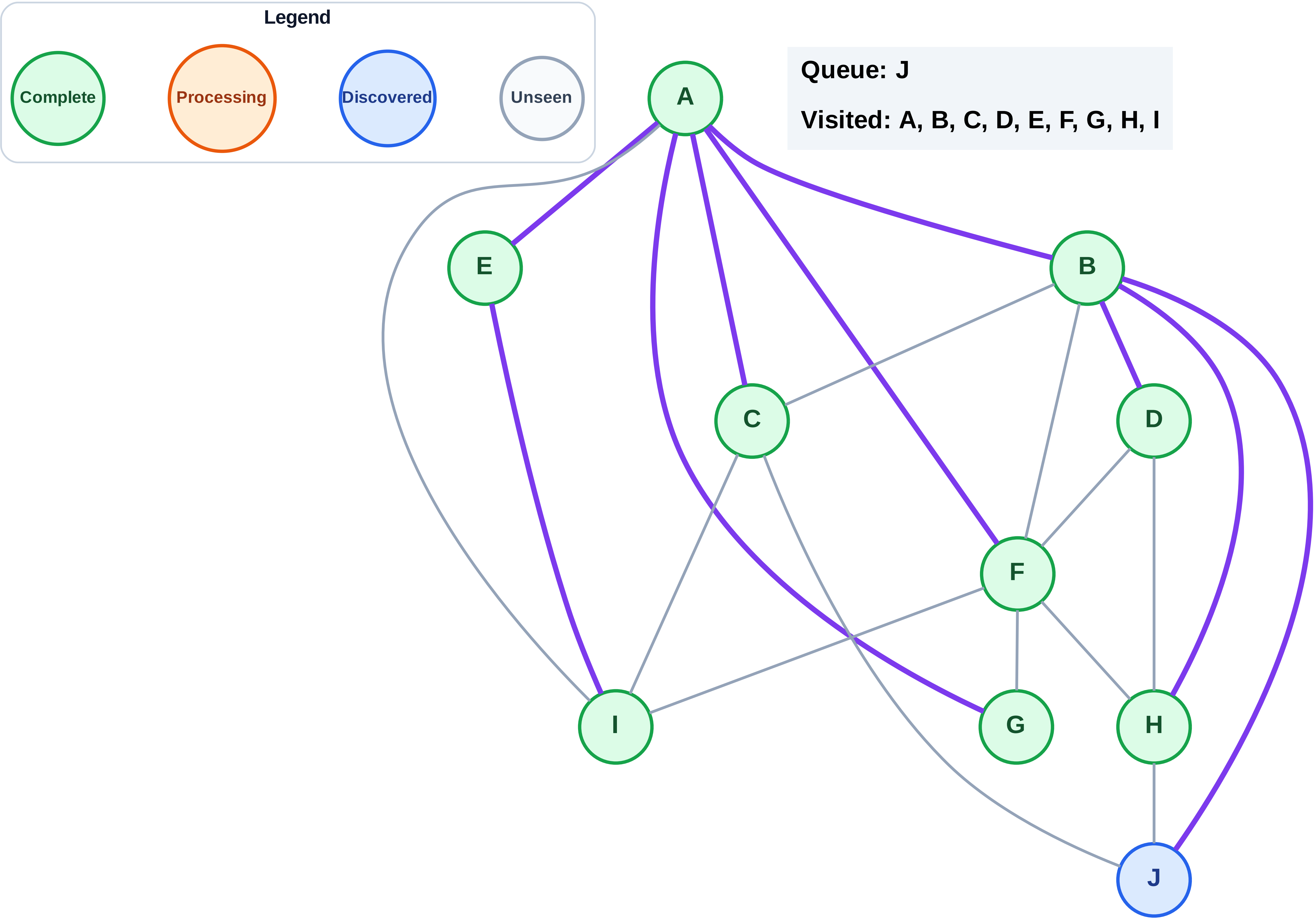
BFS Traversal

Step 18: Process node H



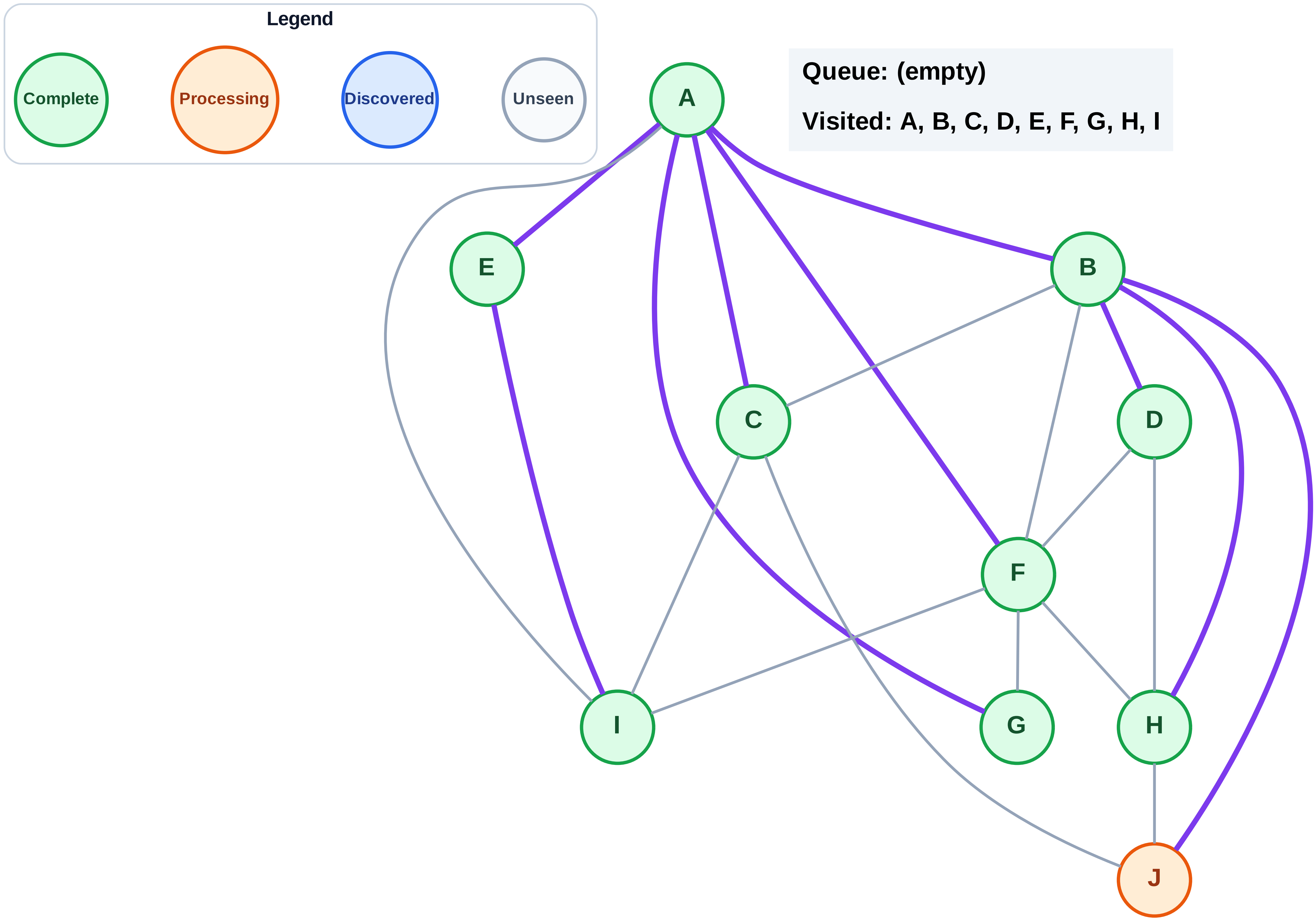
BFS Traversal

Step 19: No new neighbors from H



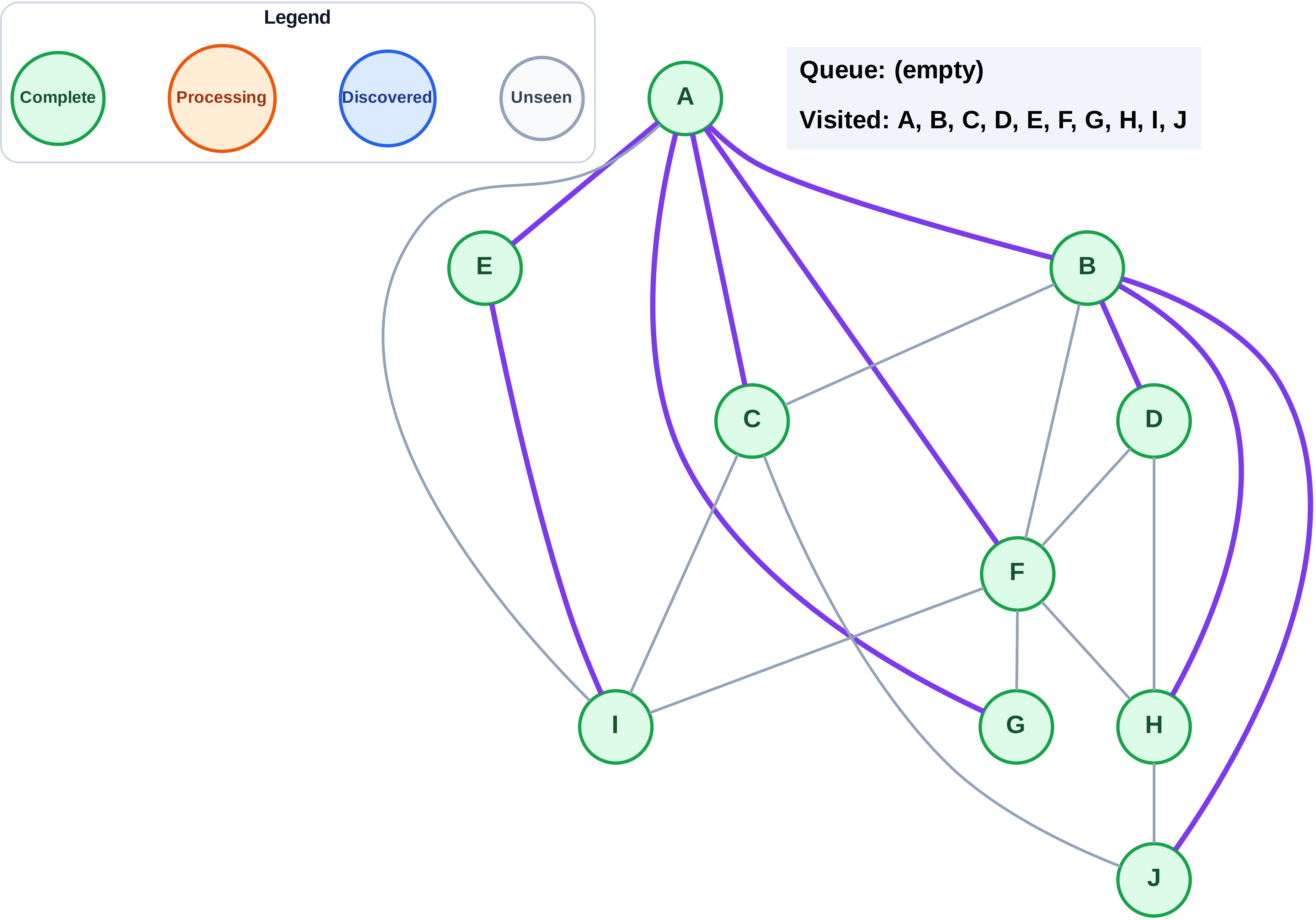
BFS Traversal

Step 20: Process node J



BFS Traversal

Step 21: No new neighbors from J

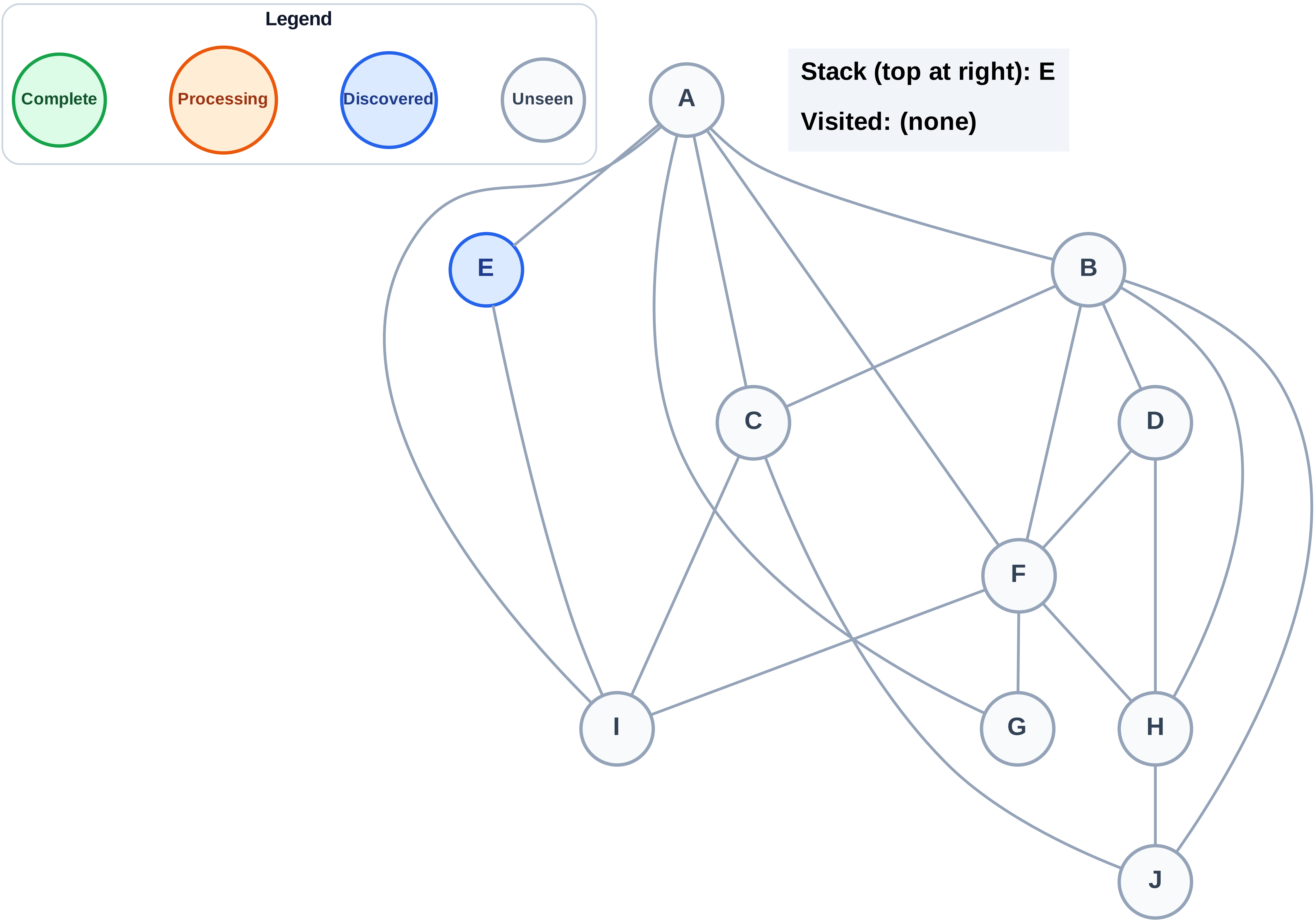


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

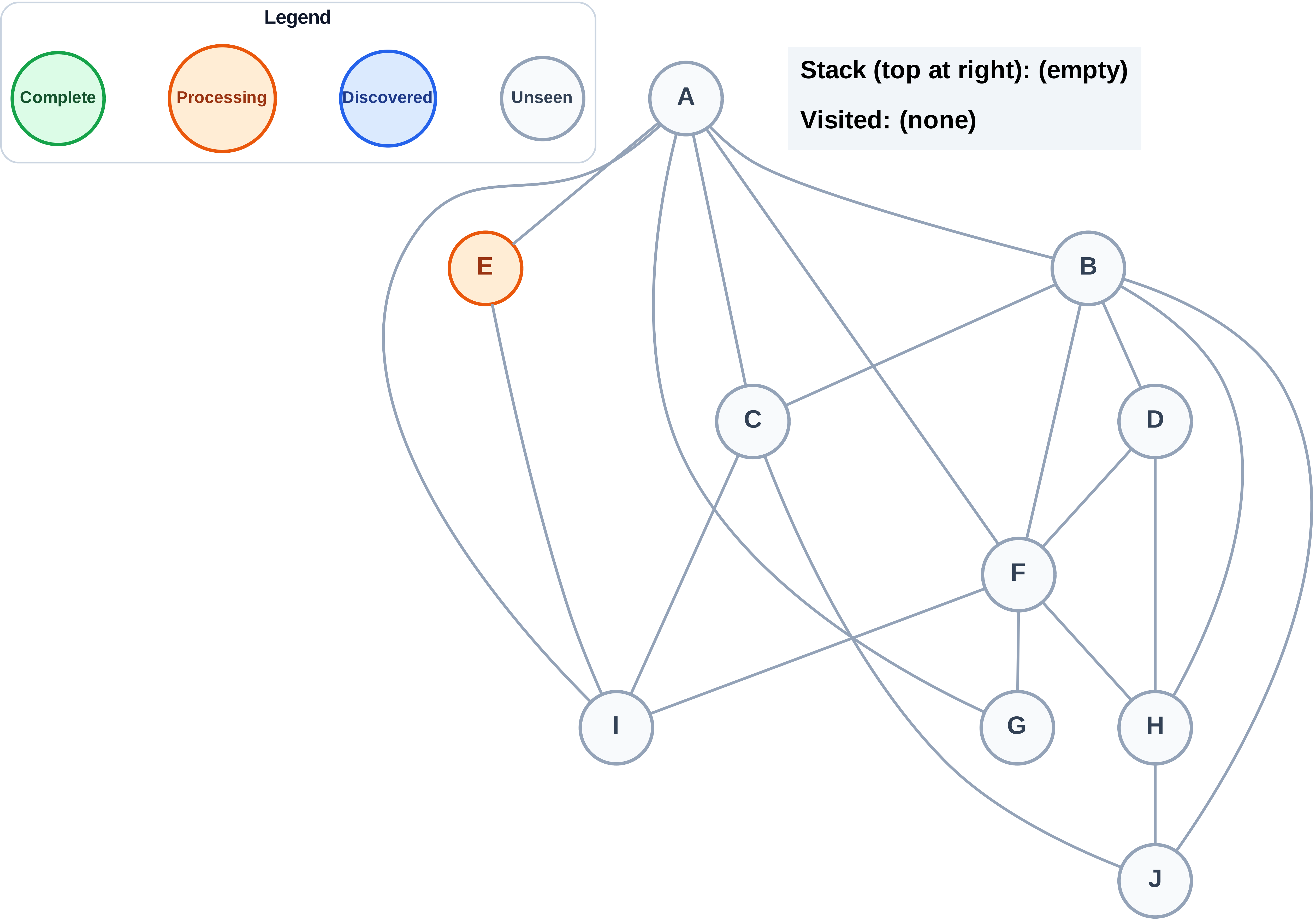
DFS Traversal

Step 1: Start at E



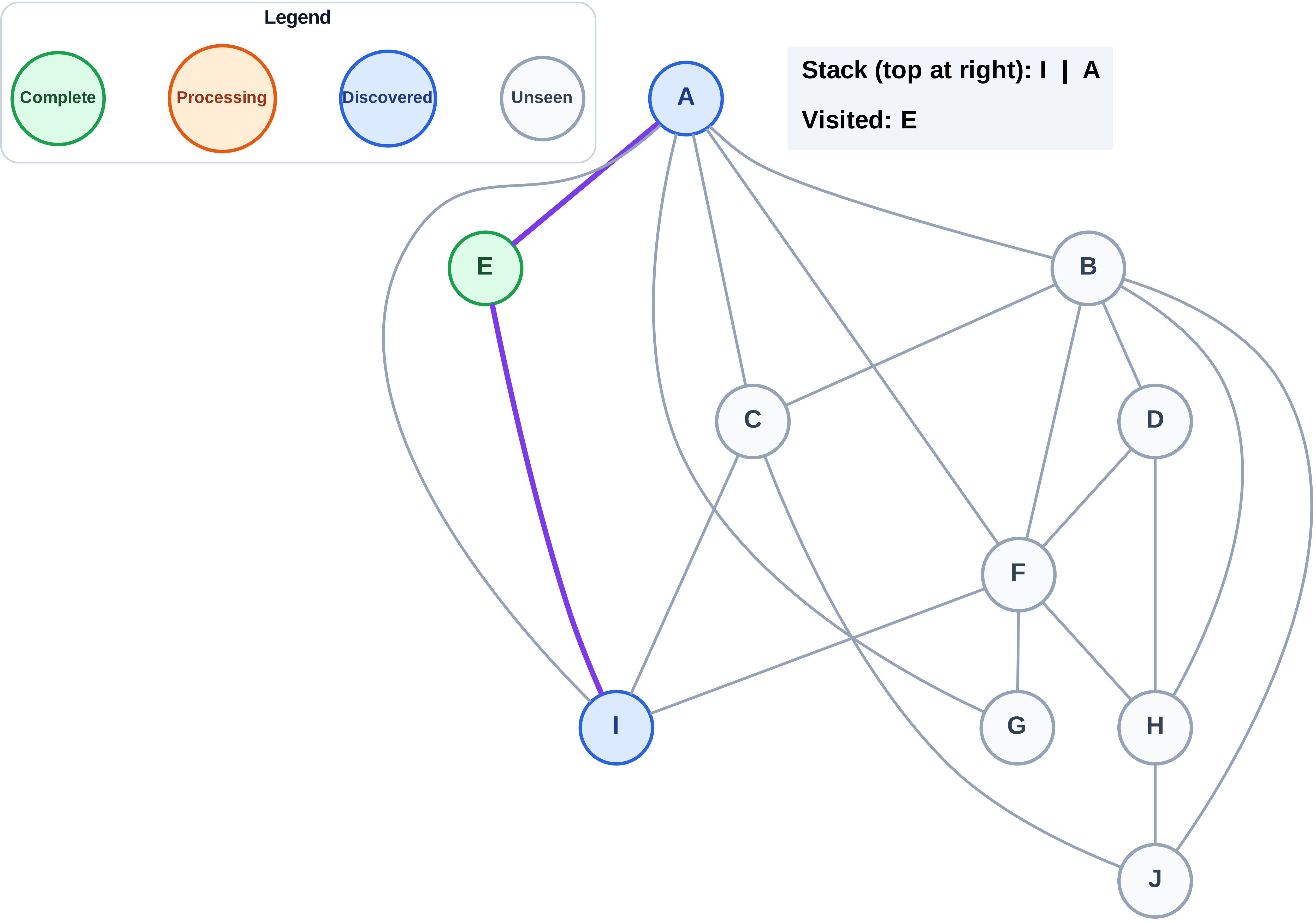
DFS Traversal

Step 2: Process node E



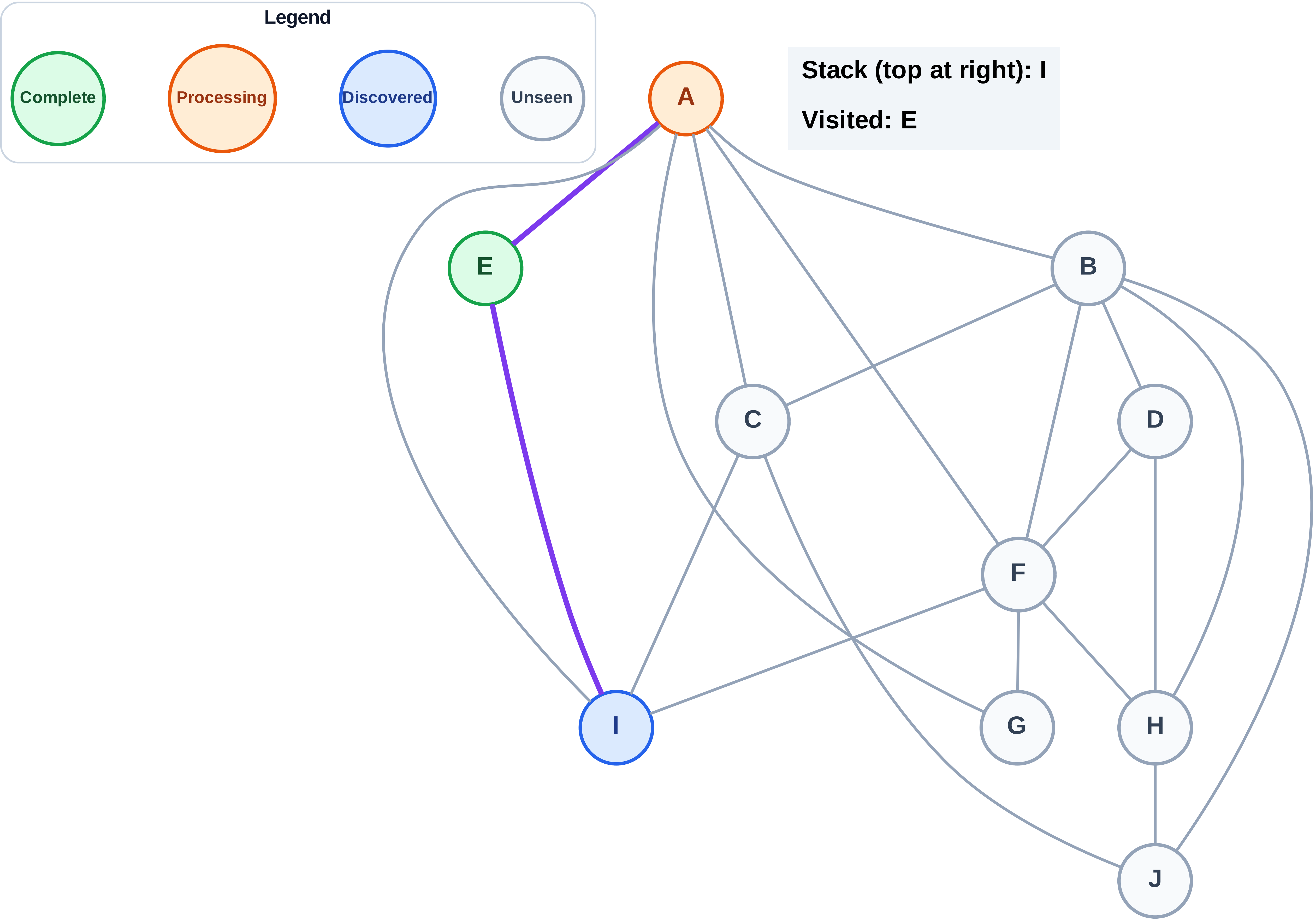
DFS Traversal

Step 3: Discover I, A from E



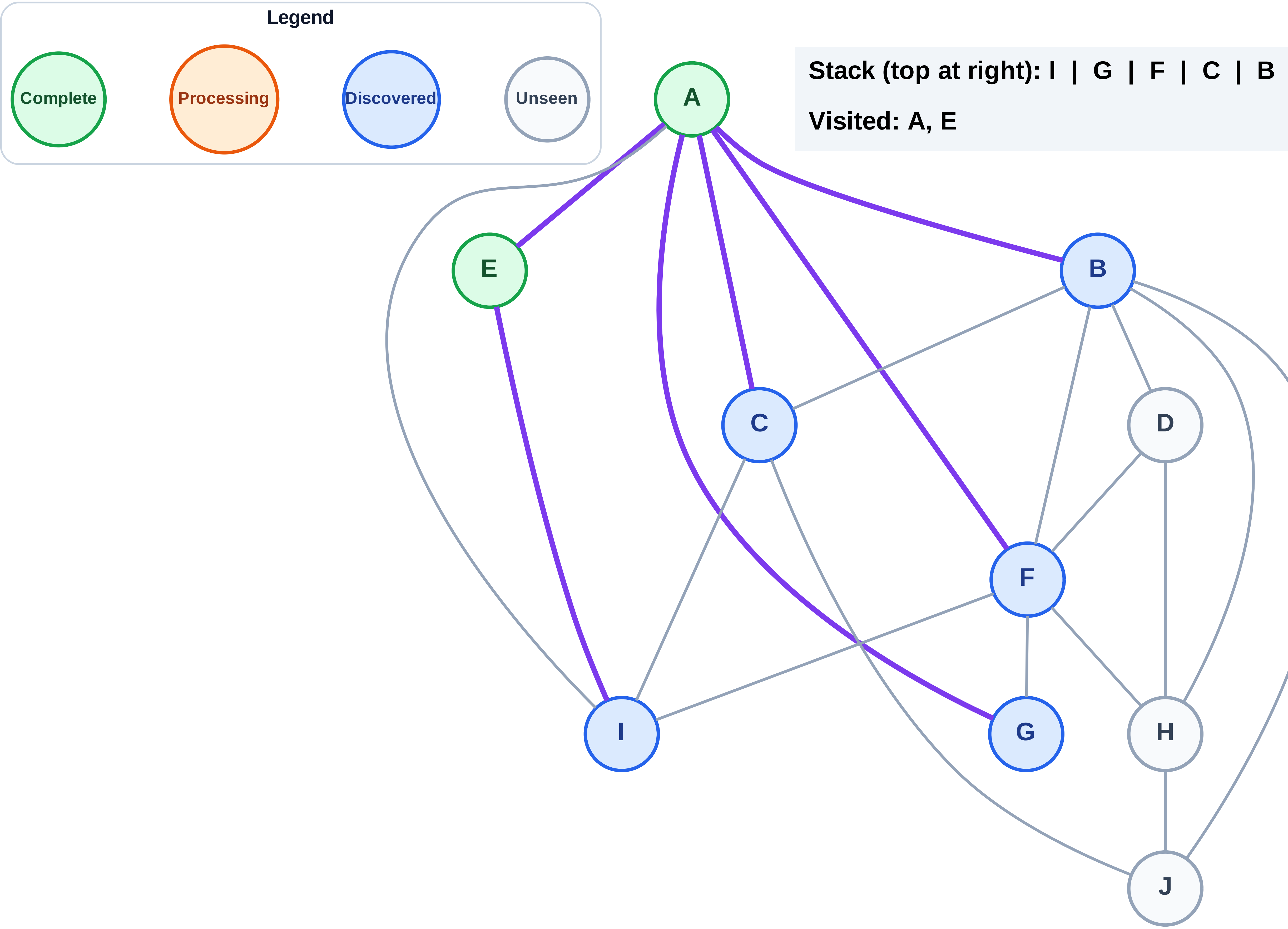
DFS Traversal

Step 4: Process node A



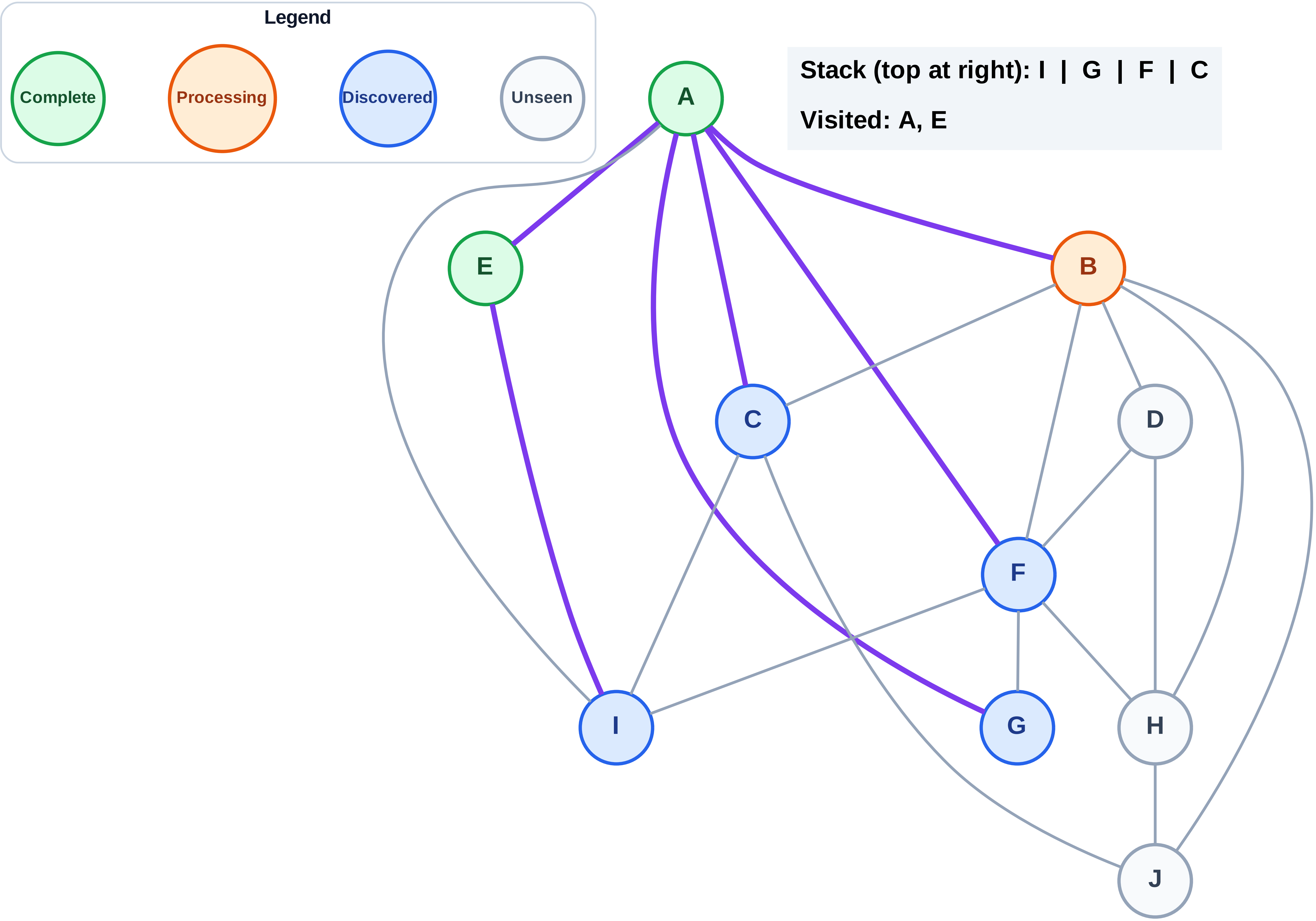
DFS Traversal

Step 5: Discover G, F, C, B from A



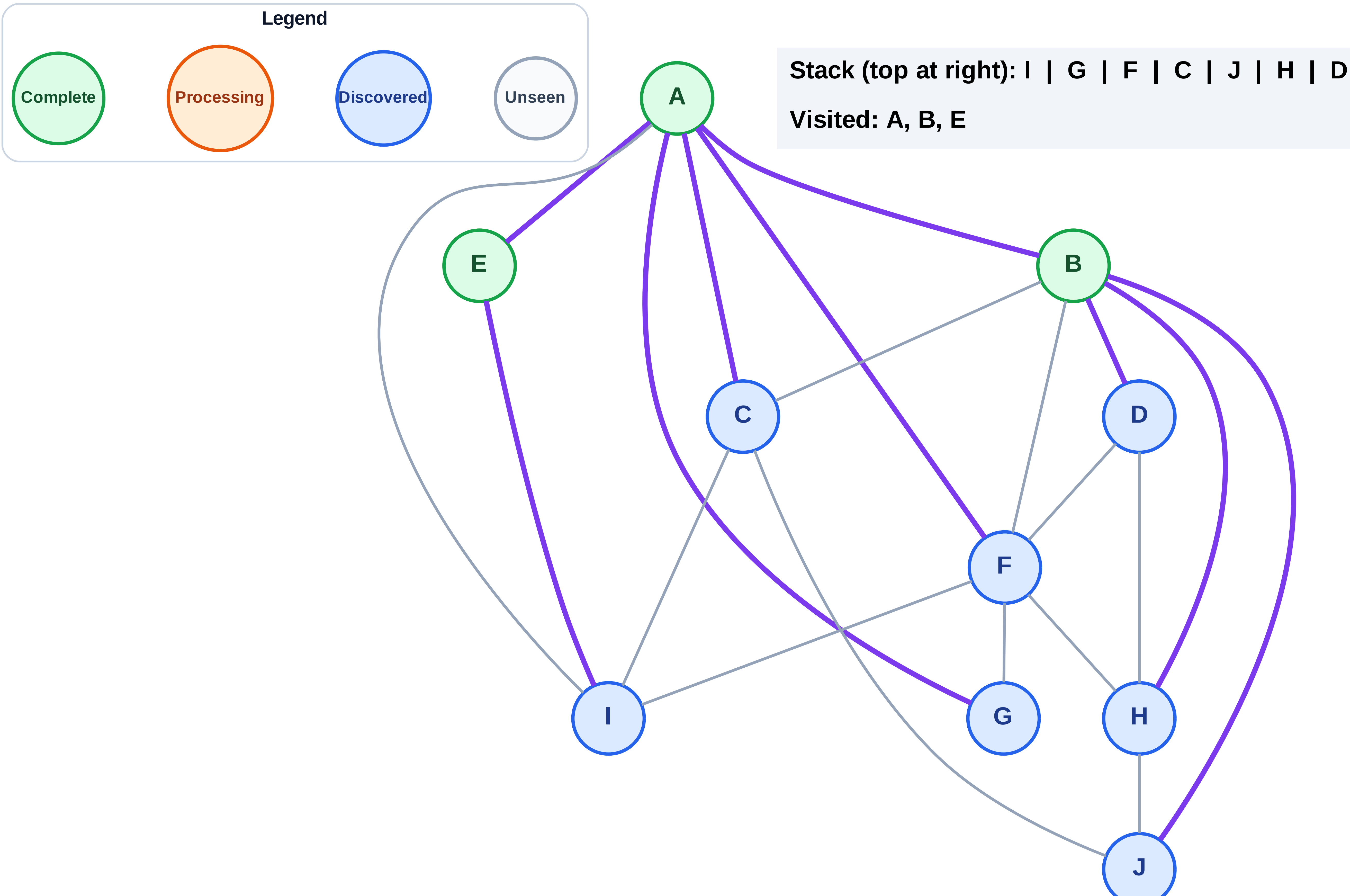
DFS Traversal

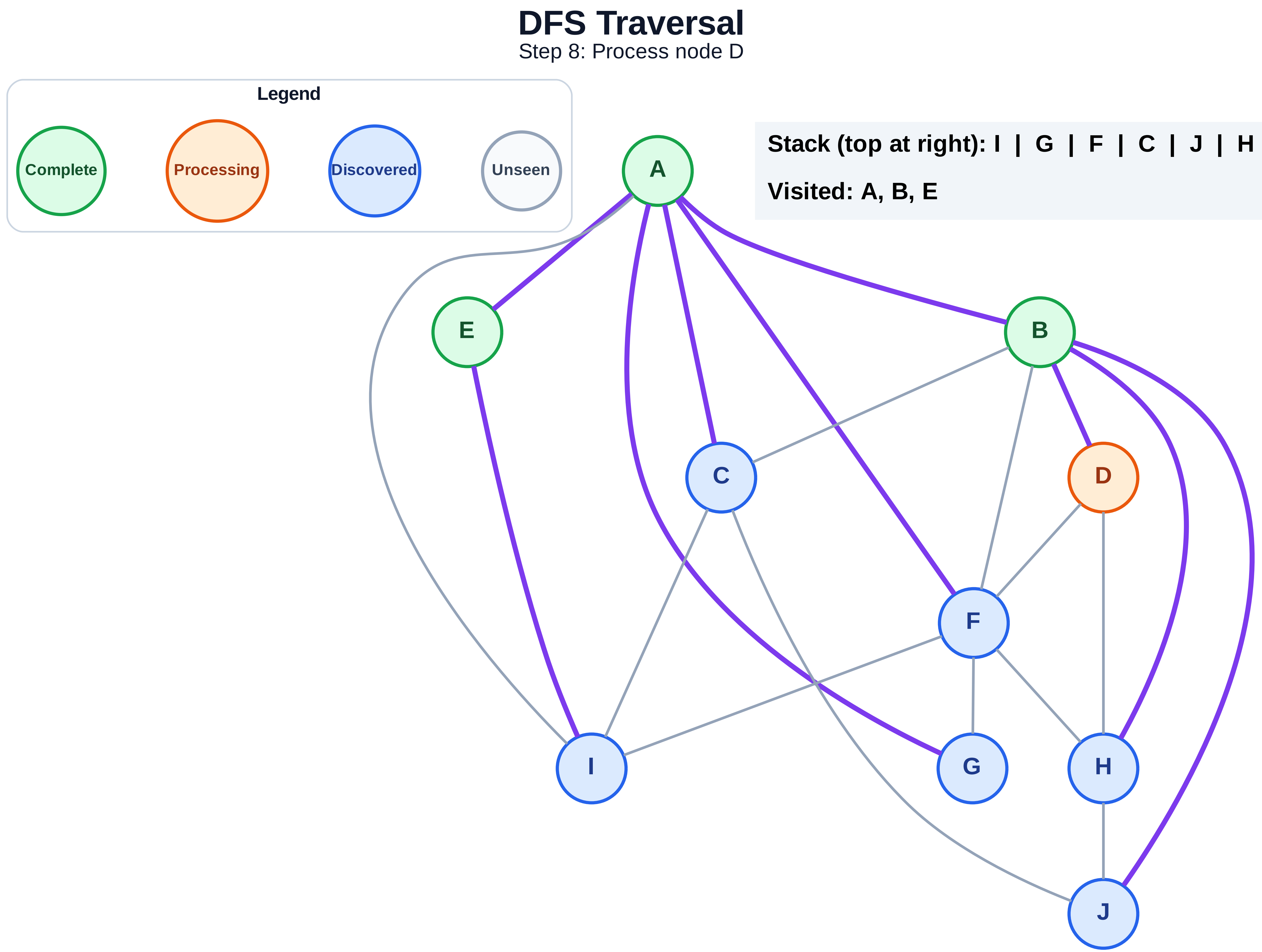
Step 6: Process node B



DFS Traversal

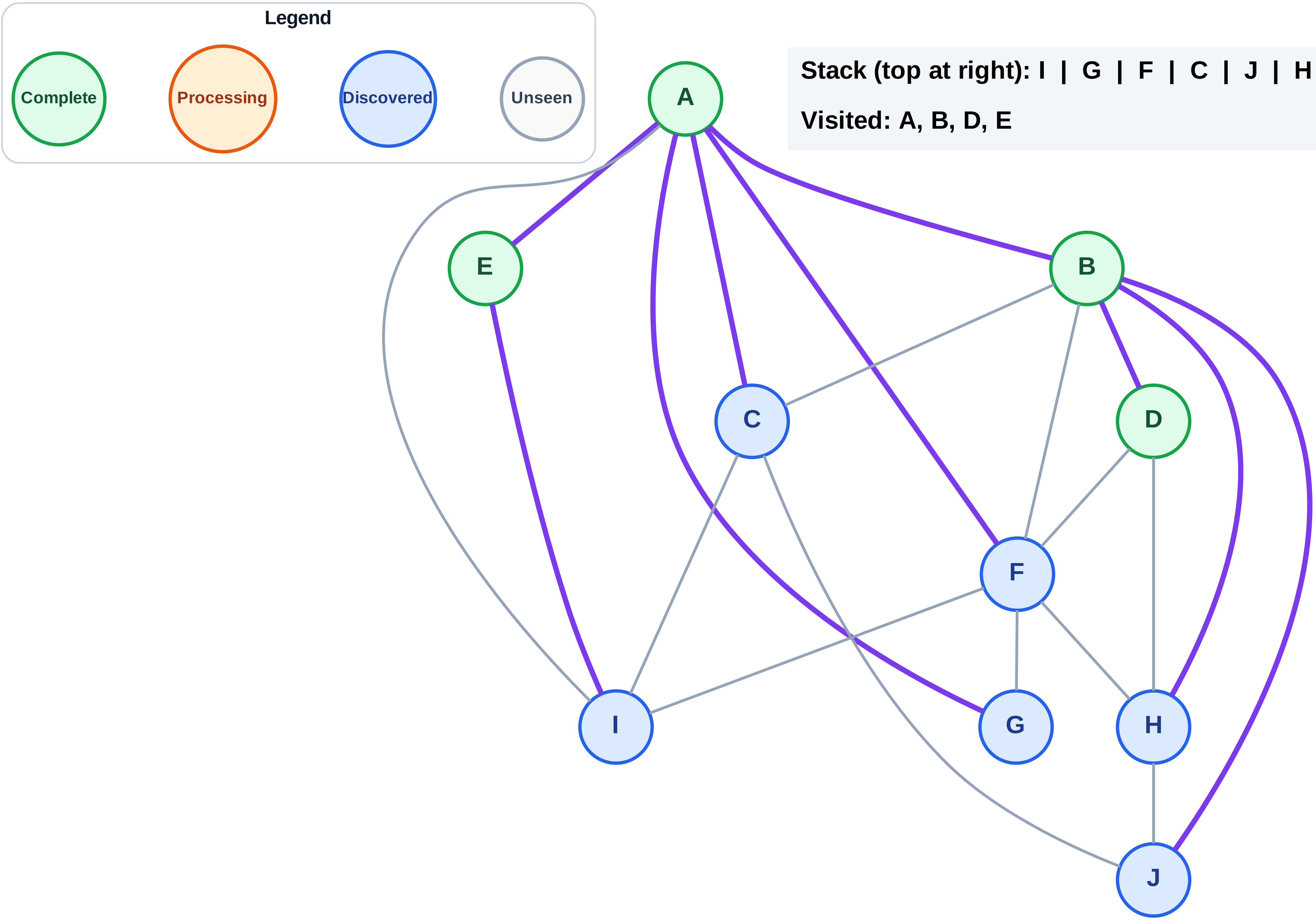
Step 7: Discover J, H, D from B




DFS Traversal
Step 8: Process node D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | G | F | C | J | H
Visited: A, B, E
A
B
C
D
E
F
G
H
I
J

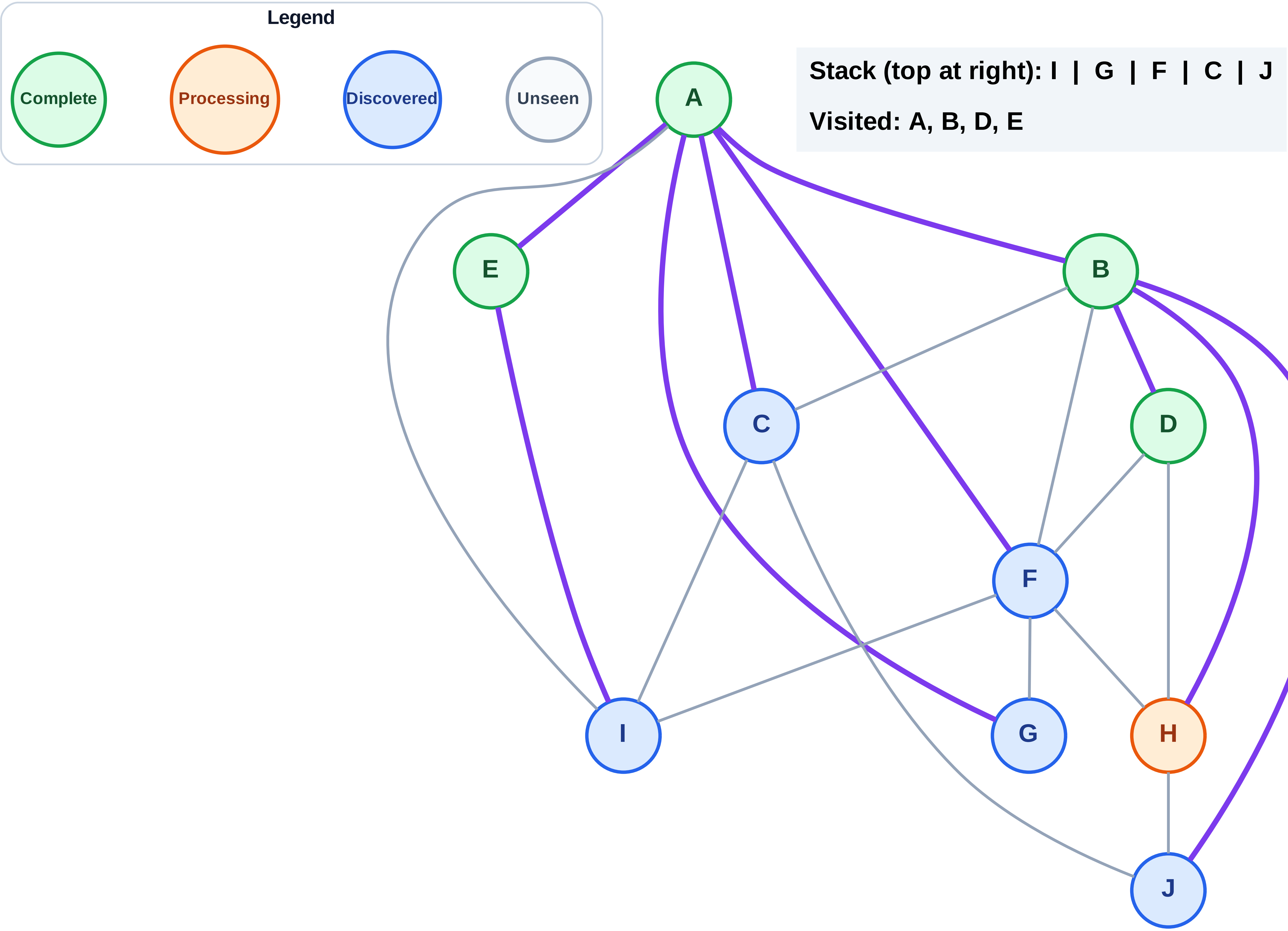
DFS Traversal

Step 9: No new neighbors from D



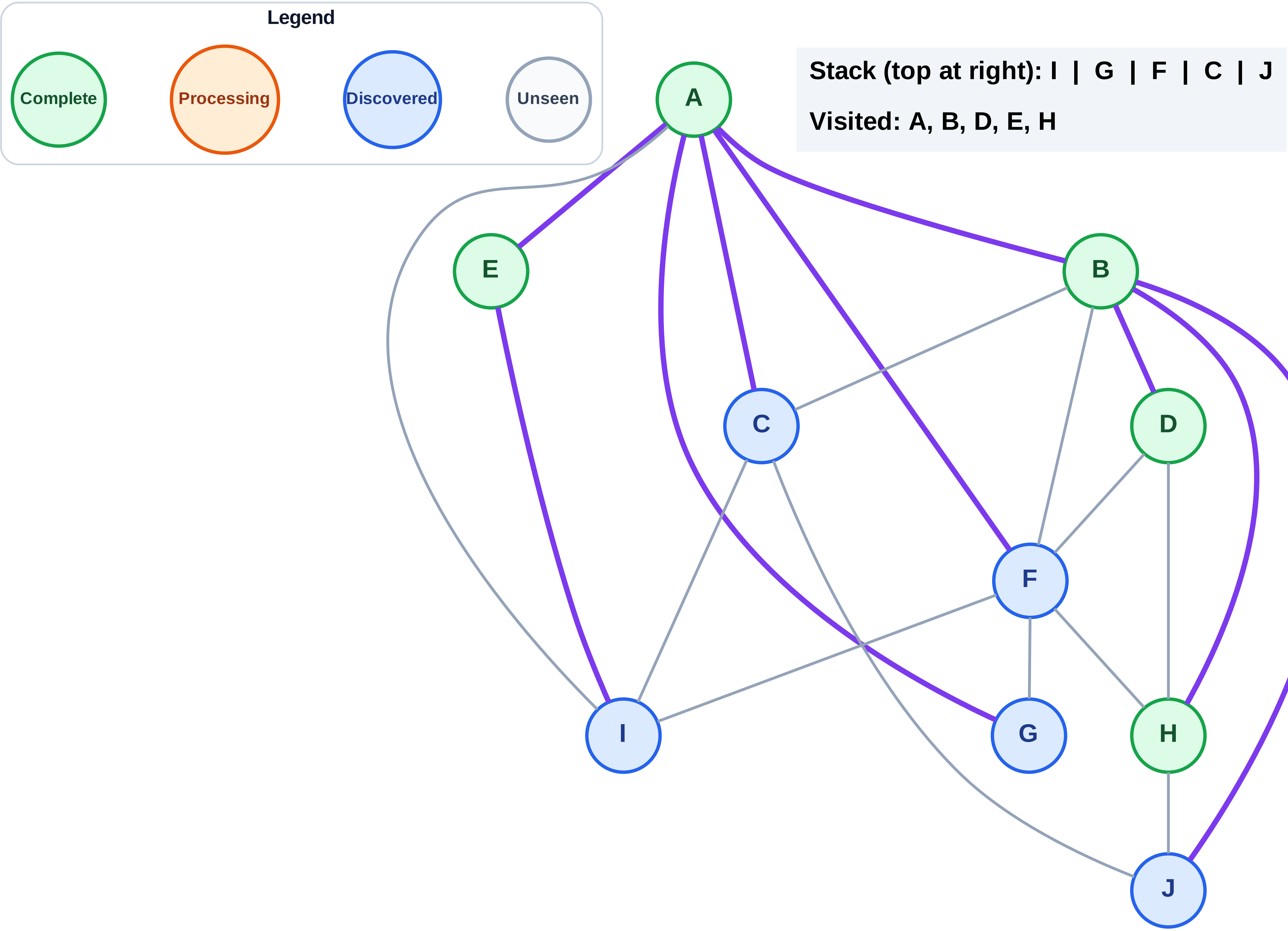
DFS Traversal

Step 10: Process node H



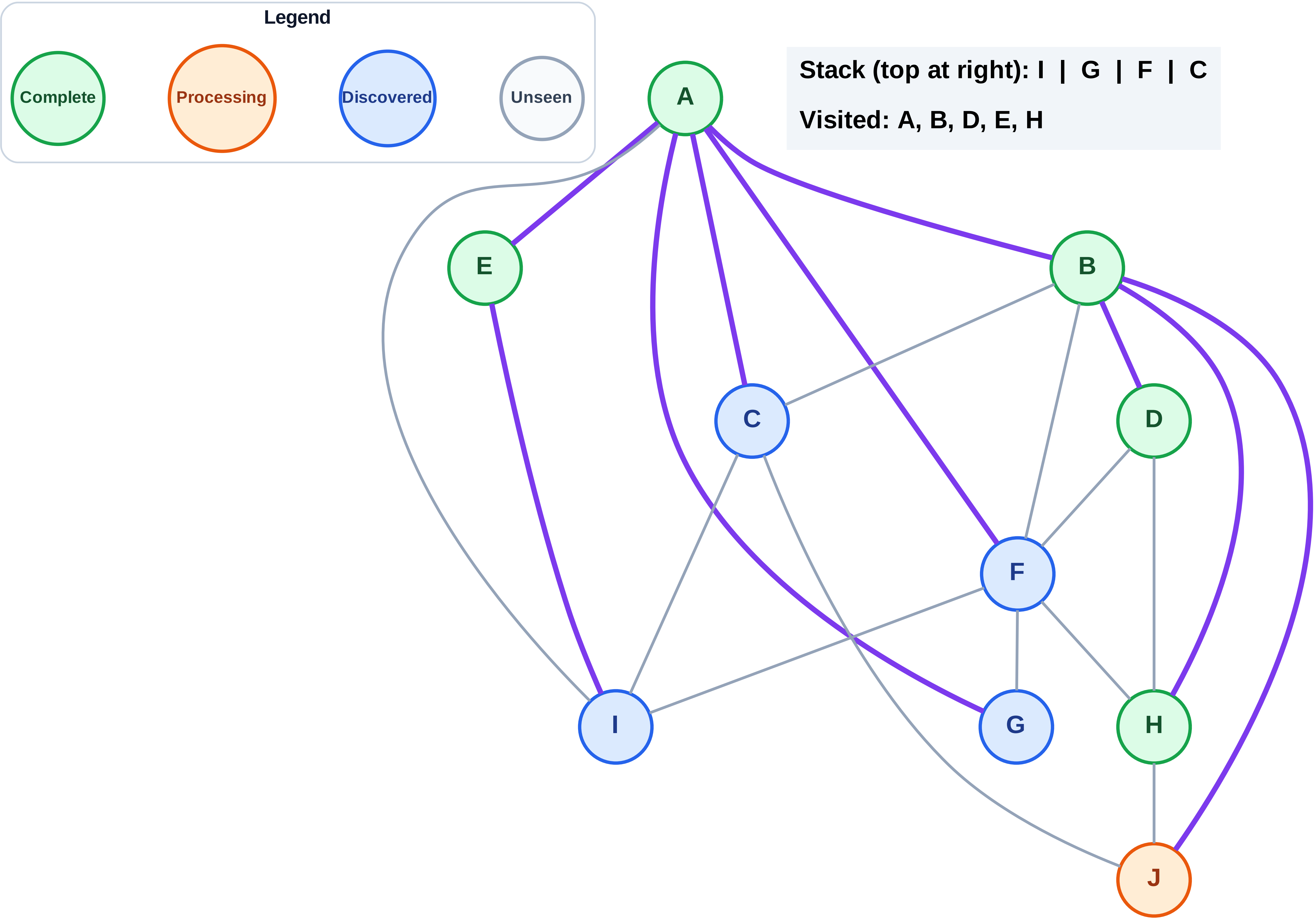
DFS Traversal

Step 11: No new neighbors from H



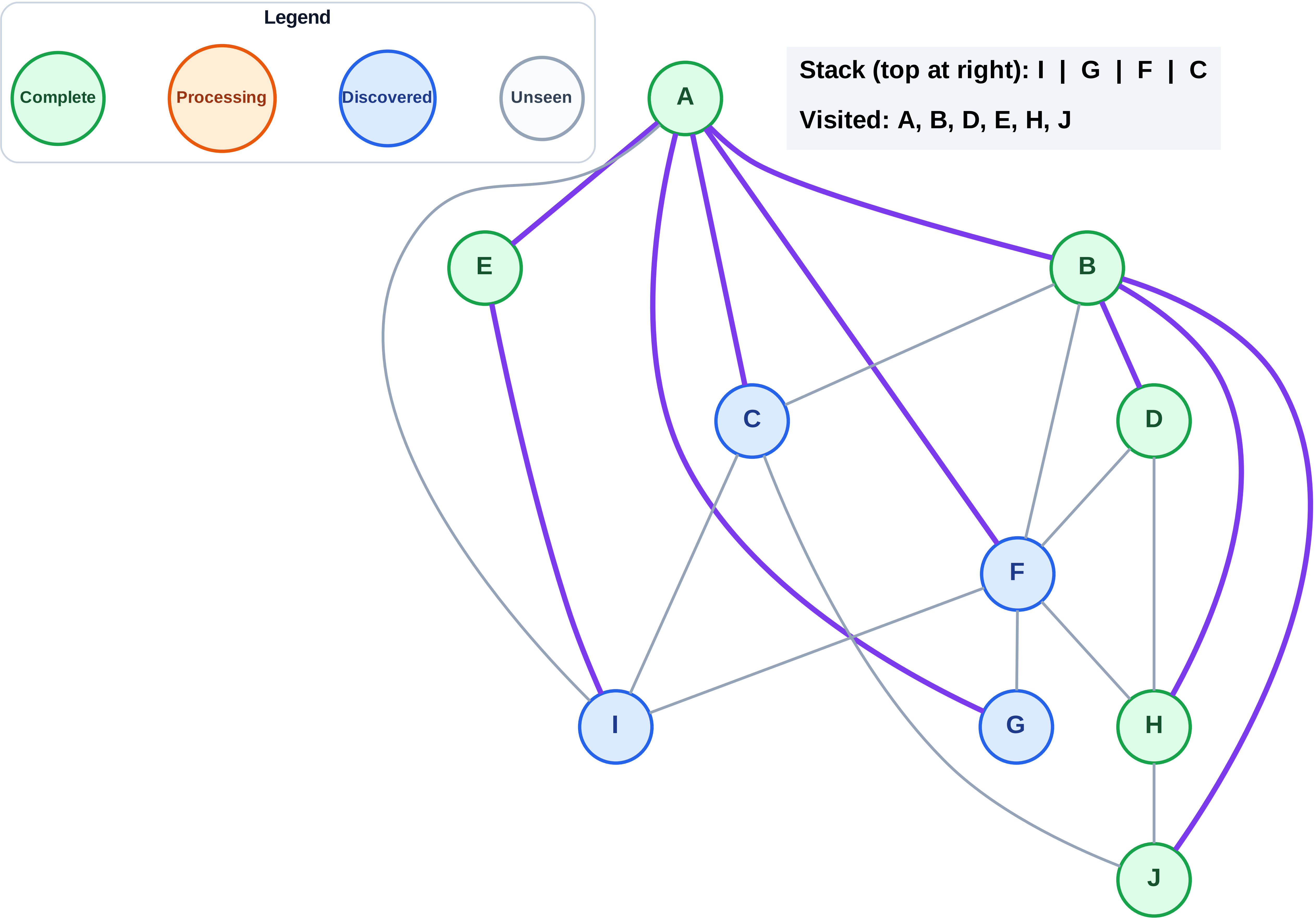
DFS Traversal

Step 12: Process node J



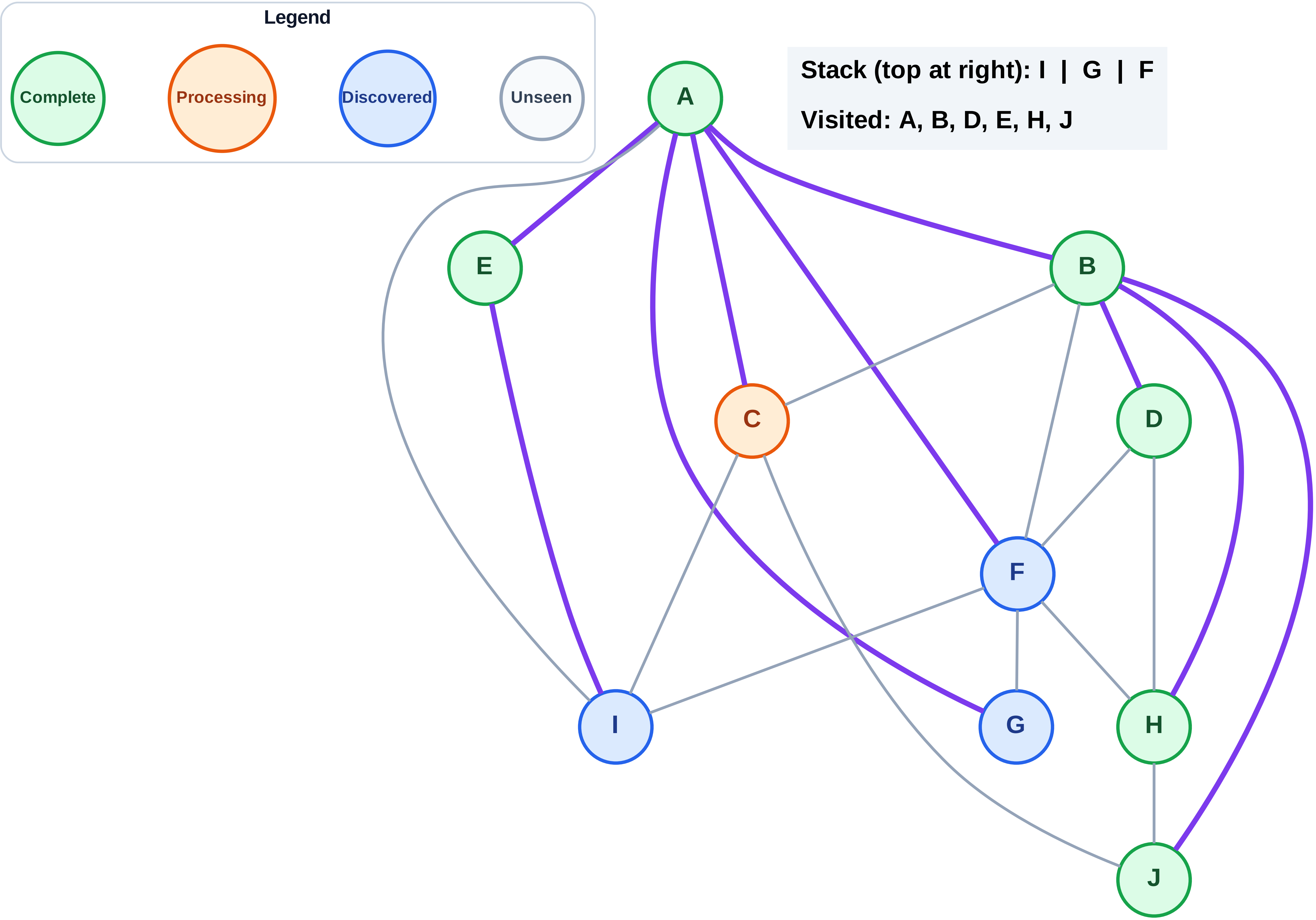
DFS Traversal

Step 13: No new neighbors from J



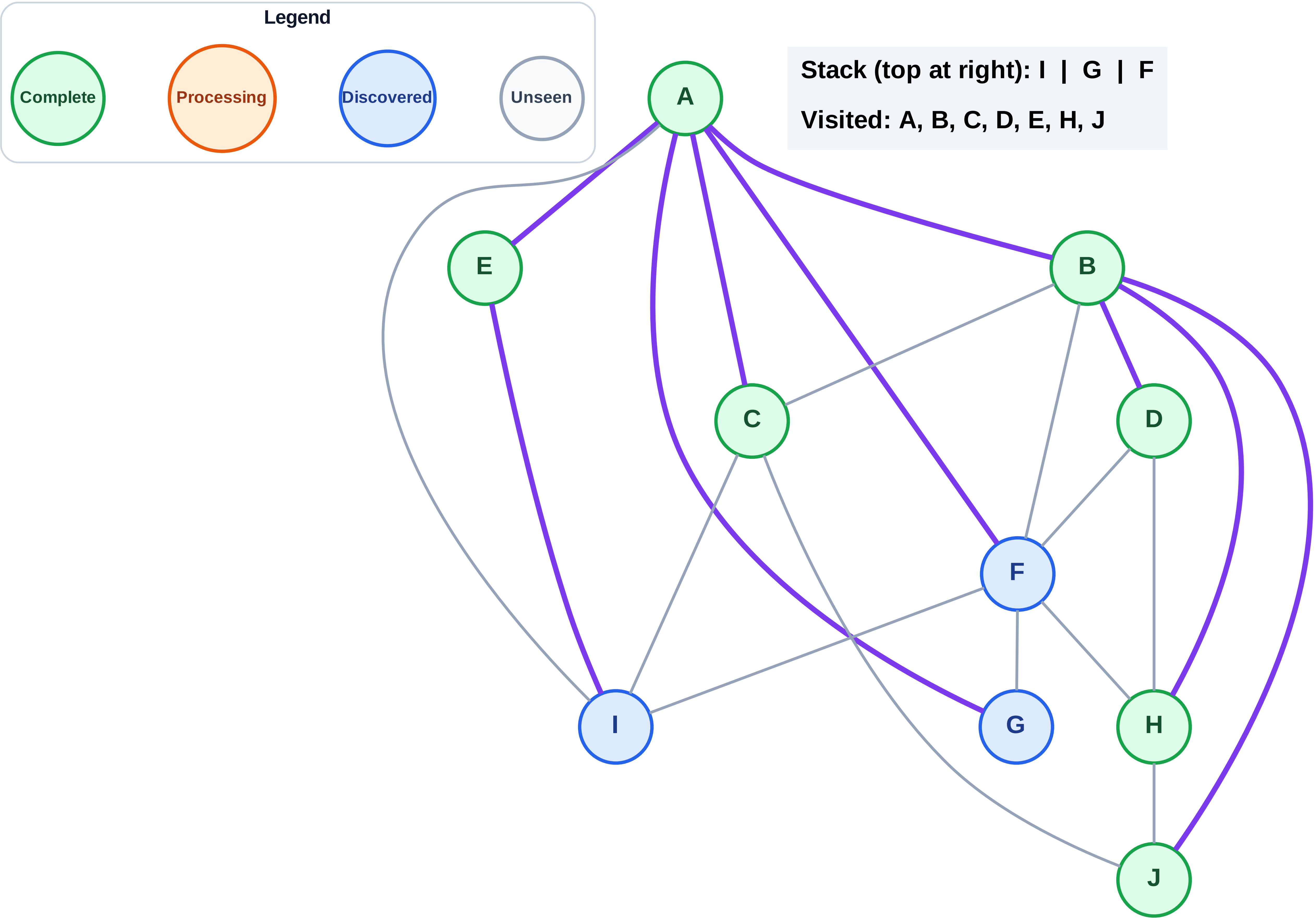
DFS Traversal

Step 14: Process node C



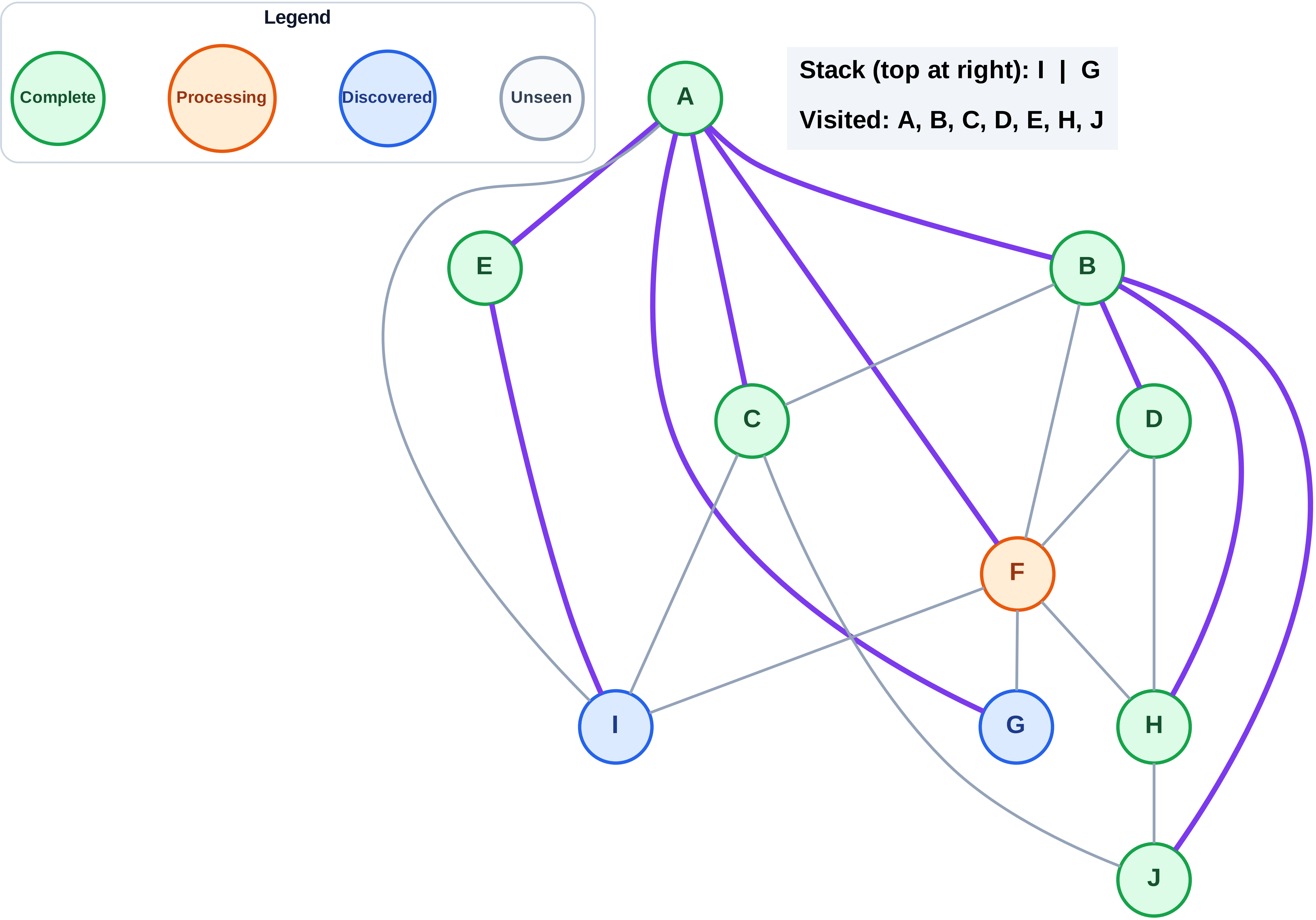
DFS Traversal

Step 15: No new neighbors from C



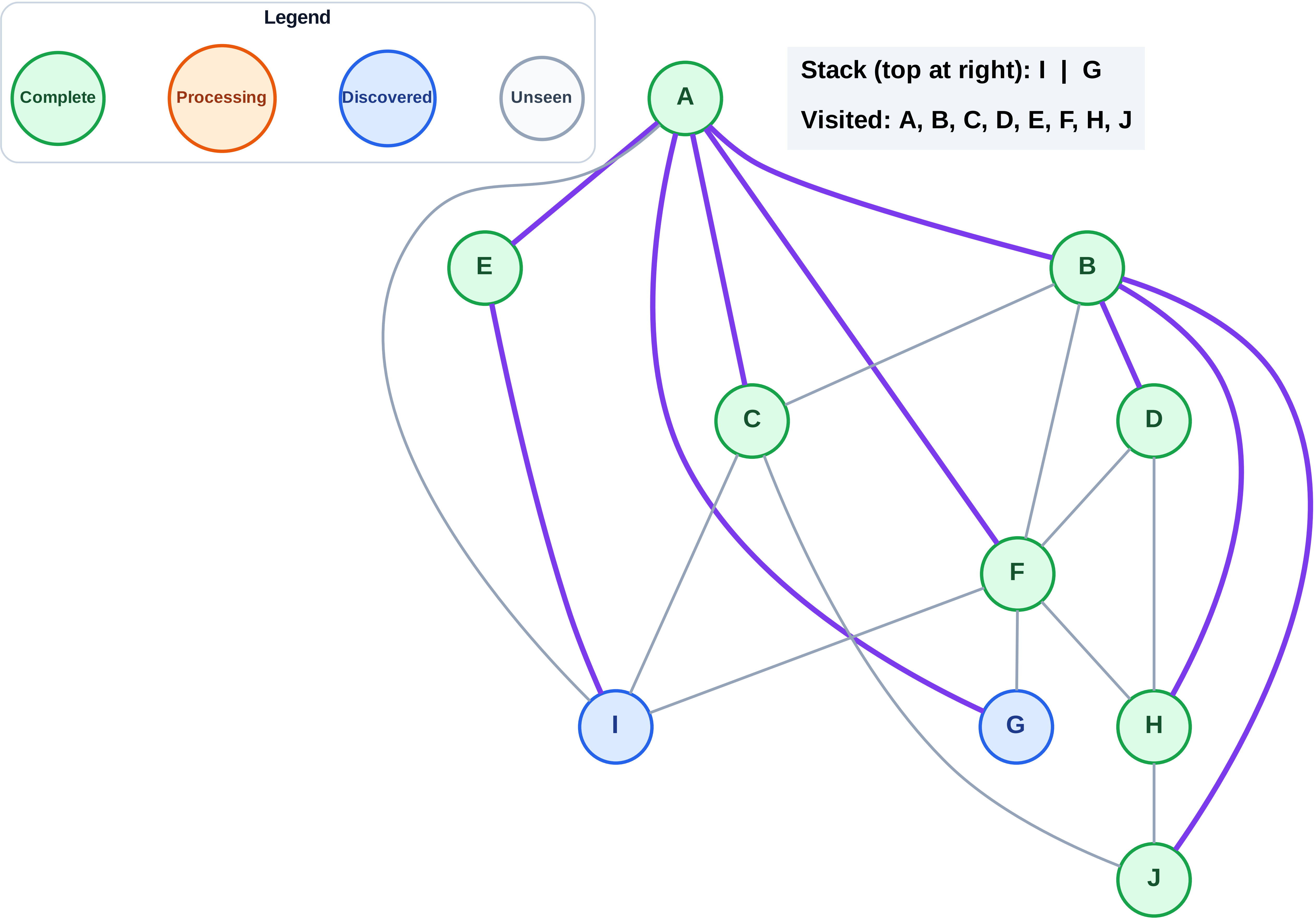
DFS Traversal

Step 16: Process node F



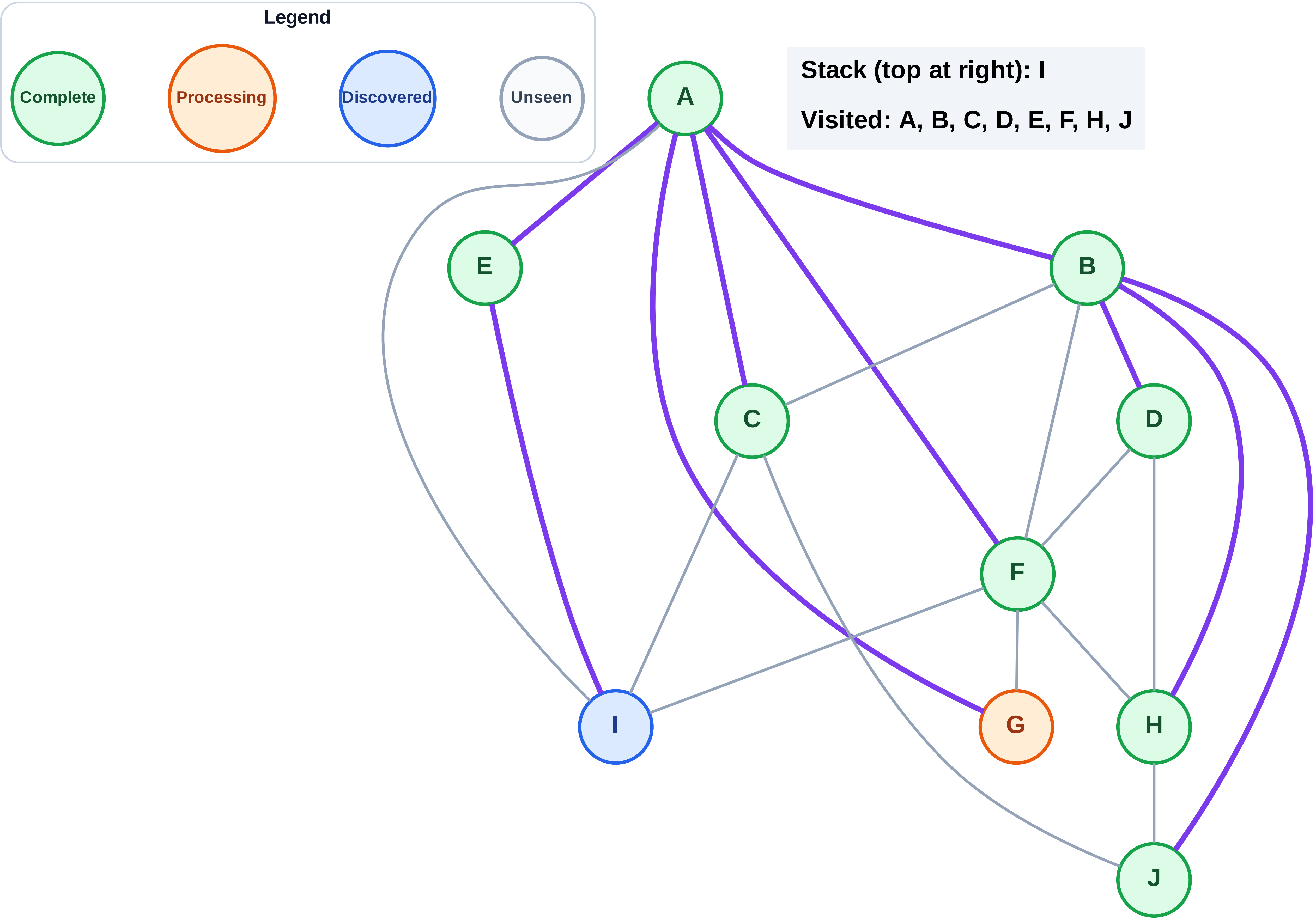
DFS Traversal

Step 17: No new neighbors from F



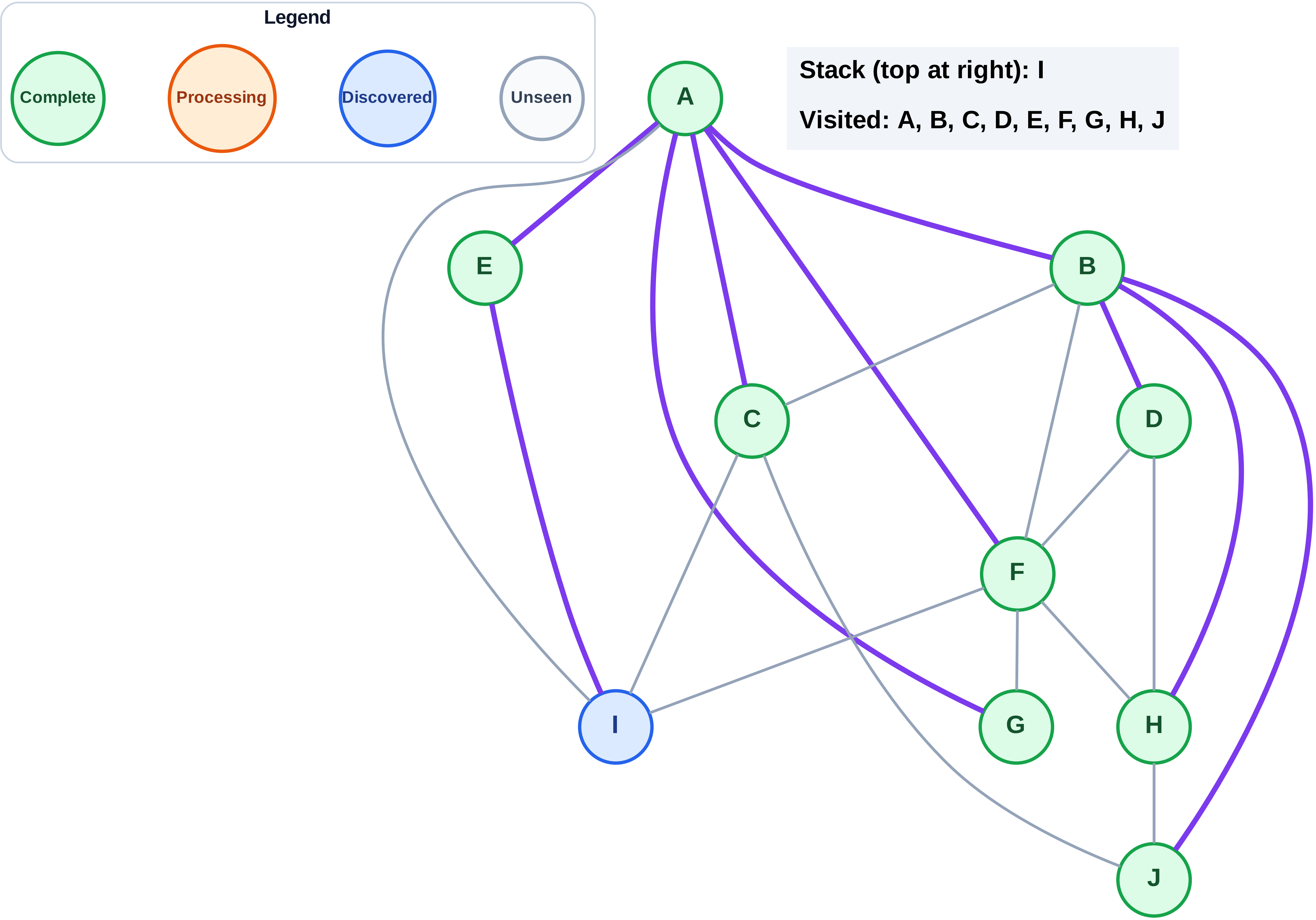
DFS Traversal

Step 18: Process node G



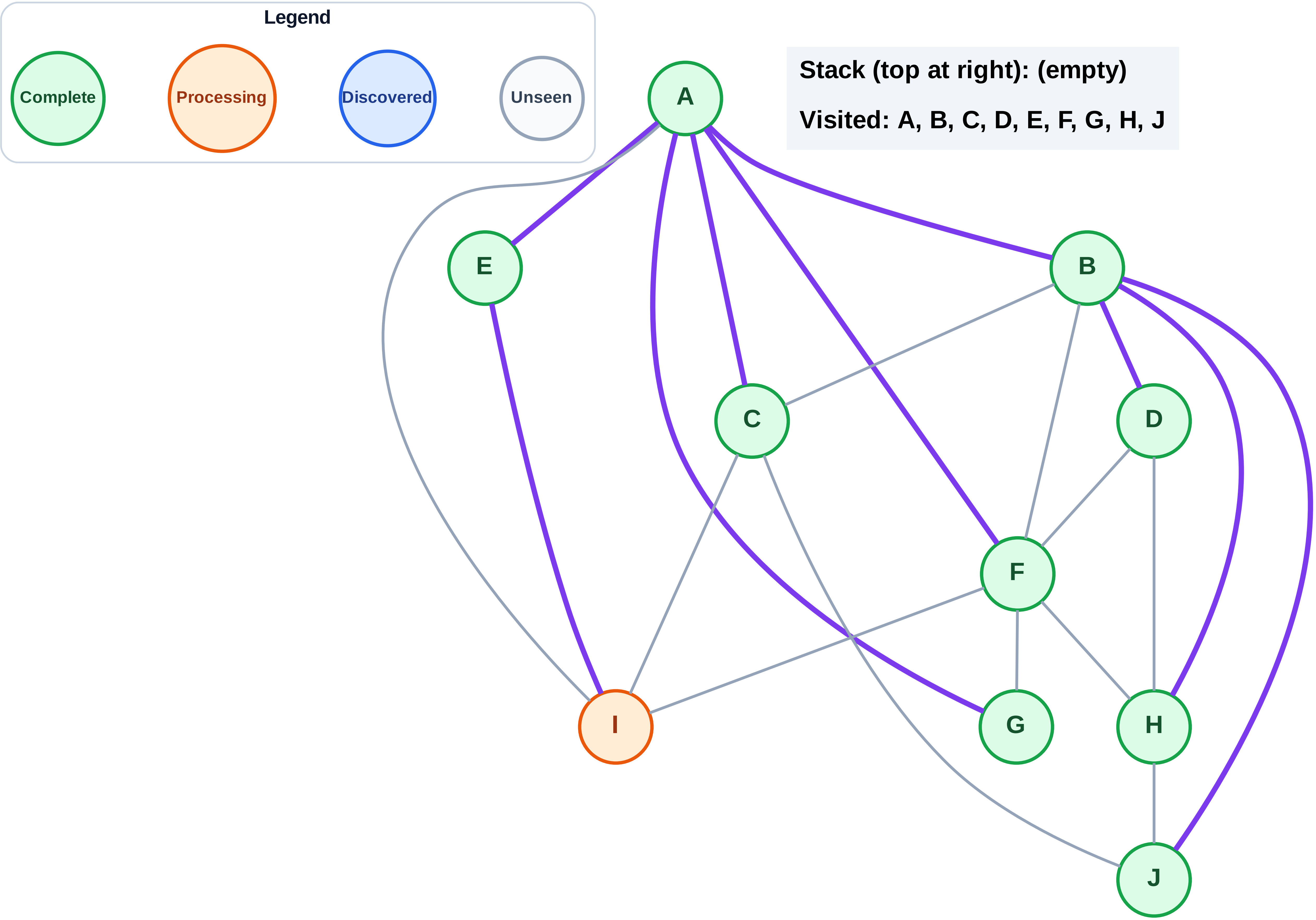
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node I



DFS Traversal

Step 21: No new neighbors from I